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# Bayesian Inference of Cancer Evolutionary Timing from Single-Cell Sequencing Data


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**Abstract**

Reconstructing the temporal sequence of mutations and genomic events during tumour expansion is a fundamental challenge in cancer data science. In this dissertation, a Bayesian model is developed and validated for the joint inference of evolutionary timings and clonal growth rates from single-cell sequencing data. The statistical inference approach uses four main factors: the number of somatic mutations accumulated along each cell clone, the occurrence of copy-number alterations (large-scale genomic duplications), the abundance of distinct cell populations, and the topology of the phylogenetic tree relating them. A hierarchical prior structure is adopted to share statistical strength across closely related clones while preserving variability across distinct subpopulations. Somatic mutations are modelled as a stochastic counting process, with intensity depending on the background mutation rate, the length of the genomic region, an event-specific coefficient, the clone-specific growth rate, and the time of evolution of the clone. The timing of copy-number alterations is inferred by partitioning the mutations mapping to each affected genomic region into those arising before and after the event, exploiting the discontinuity in the mutation accumulation process induced by the copy-number change as a temporal signal.

The inference framework is assessed through simulation studies of increasing complexity. Under a simple generative scenario the model recovers ground-truth timings and growth rates with good accuracy. A more demanding evaluation is carried out using synthetic data from ProCESS, a cancer evolution simulator, which allows examination of how deviations from the modelling assumptions propagate into the inference outcomes.

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# Introduction

Cancer is an evolutionary process. A tumour develops from a single founding cell and grows through repeated cycles of cell division, and at each division somatic mutations arise. Most of these mutations are neutral, leaving the cell's growth and fitness unchanged; a few give a selective advantage and seed new subclones that expand on their own. A tumour observed at the moment of sequencing is therefore a mixture of clones, each carrying its own history of mutations and each having grown at its own pace.

Reconstructing that history, meaning which events happened, in what order, and how quickly each clone expanded, is one of the central problems of cancer data science. The timing of mutations and copy-number changes, and the speed at which clones grow, say something about how a tumour has progressed and, possibly, about how it will behave, so a reconstruction that is both accurate and aware of its own uncertainty is useful scientifically and clinically. The difficulty is that the process is observed only once: a single snapshot has to be used to infer something that unfolded over time. This makes careful statistical modelling, and a clear statement of what can and cannot be recovered from one observation, essential.

The data for this task come in two main forms, which differ in cost, resolution and the kind of information they give. Bulk sequencing is still comparatively cheap, but it returns the average signal from a mixed population of cells and so loses information about the sizes of the clonal subpopulations, which limits how well the diversity of the tumour can be resolved. Single-cell sequencing instead exposes tumour heterogeneity in much greater detail and gives information about the subclonal populations that bulk experiments cannot see. This resolution comes at a price: single-cell reads are typically noisy and harder to interpret, so the data need careful processing.

A first step in analysing such data is to reconstruct the phylogenetic tree relating the sampled cells. Several methods do this, for instance *Pharming* (Weber et al. 2026) and *BiTSC*$^2$ (Chen et al. 2022), but a tree inferred from data is an estimate, not a ground truth:

it fixes the ordering constraints among events without, by itself, telling us when those events occurred or how fast the clones grew. This is the gap the present work addresses. Taking the inferred phylogeny as its starting point, the thesis develops a Bayesian hierarchical model that estimates the timing of mutation and copy-number events together with the clone-specific growth rates. The model ties several signals to the same latent timeline: the number of mutations accumulated along each branch, the way mutations in a copy-number region split into those arising before and after the alteration, and the relative abundance of the clones. It places a hierarchical prior on growth, so that closely related clones share statistical strength while distinct ones remain free to differ.

A comparable amount of attention is given to showing that these inferences can be trusted. The model is validated in stages of increasing difficulty. It is first fitted to data drawn from its own generative process, where parameter recovery and simulation-based calibration check that the computation is faithful and the reported uncertainties are calibrated; it is then tested on synthetic data from ProCESS, a spatial tumour-evolution simulator, which shows how the inference behaves in a more realistic setting. This staged validation is itself part of the contribution of the thesis.

The dissertation is organized as follows. Chapter 1 reviews the biological background and the sequencing technologies, and frames the inference problem in statistical terms. Chapter 2 develops the models: a first, simpler model that treats the growth rates as known and infers only the event times, and a second that estimates the growth rates as well. Chapter 3 presents the simulation study: the ideal data-generating process, the calibration analysis, and the experiments on ProCESS data. Chapter 4 discusses the findings, their limitations, and directions for future work. Two appendices collect the mathematical derivations underlying the model and the details of its computational implementation.

# 1. Biological and Statistical Background

## 1.1 Biological Framework

Cancer growth can be thought of as a dynamic process that unfolds within a population of cells. A tumor usually starts with a mutation of one cell, which subsequently starts dividing. During the subsequent cell division events, the initial mutant cell produces a lineage of cells, which grow into a subclone of descendants. During the growth of this lineage, cells accumulate somatic mutations and these are passed onto descendants. Similar to other evolutionary phenomena, somatic mutations bring diversity to the system, and the outcome depends on how various clones are fitter than others, expanding into large populations, staying rare, or dying off.

A tumor is an inherently stochastic process. Genetic mutations occur spontaneously either during the process of DNA replication or following any DNA damages and imprecise repair mechanism. Moreover, there is no guarantee that a potentially successful mutation will make a cell live longer or multiply: a cell carrying the mutation might simply die or remain small. Thus, when observing the final composition of a tumor, it is impossible to disentangle what exactly happened to individual cells in terms of biology and stochastic factors linked to cell proliferation and death, and mutation acquisition and extinction.

In this setting, mutations are often classified as passengers or drivers. Passenger mutations are neutral mutations that do not change cell biology and do not affect its proliferation or fitness. Yet, they are very useful as ancestry indicators: cells with identical passenger mutations share ancestry. Driver mutations represent changes that affect cell biology, such as growth, replication, survival, or other functions relevant to cancer evolution. These mutations may provide some selective advantage to a mutated cell, driving the formation of a subclone.

As a tumor grows, its evolutionary history can be represented with a phylogenetic tree. The tree is rooted at the first tumor cell that emerged and contains information about

accumulated mutations and the formation of clones. Here, clones correspond to groups of cells that share identical sets of mutations, and a subclone appears in the case when a cell within a clone acquires another set of mutations and grows. In particular, at the moment of sequencing, only those cells that survive are observed, and it is required to infer the evolutionary process behind a given set of observed genotypes.

Another fundamental parameter in cancer is mutation rate. This rate can be defined, for instance, as the frequency of mutations appearing per cell division. Under some stochastic assumptions a specific number of mutations can be used as a proxy to measure branch length: the more mutations there are along a branch of a phylogenetic tree, the older the event, and vice versa. However, mutations do not necessarily accumulate uniformly along a branch.

Another type of somatic events in tumors is copy-number alterations. While a normal diploid region consists of a copy of two alleles, one from the mother and another from the father, cancer genomes can have gains, losses, or duplications of certain genomic fragments. In this work, we consider three main types of copy numbers, illustrated in Figure 1.1: (2:1), (2:2), and (2:0). Copy-number (2:1) denotes trisomy, which means that one allele is in double copy while the other allele is only in one copy, making the total number of copies equal to three. Copy-number (2:2) denotes tetrasomy, which could result from WGD where both alleles are duplicated, making the total number of copies four. Finally, copy-number (2:0) denotes copy-neutral loss of heterozygosity (CNLOH), where an allele gets deleted while the remaining allele gets duplicated, resulting in total copies of two but one allele being completely deleted.

Another fundamental concept is clone growth rate. Growth rate defines how fast a specific clone grows over time. A faster-growing clone is able to reach a higher frequency of its occurrence within the tumor population. As a result, the final frequencies of clones depend not only on their age but also on their growth rates. Specifically, early clones have had more time to grow, yet their growth rates could be relatively slow. On the other hand, newer clones are younger, yet they can have higher growth rates and therefore be present in higher share.

It becomes clear now that the analysis of tumor evolution requires probabilistic models to account for multiple aspects of the process. Number of mutations gives information about the timing of different events, clone proportions in the sampled cells give information about their expansion dynamics, and phylogenetic trees show ordering between the clones. While the constructed phylogenetic tree should not be treated as exact ground truth, it remains an effective way to model the tumor process.

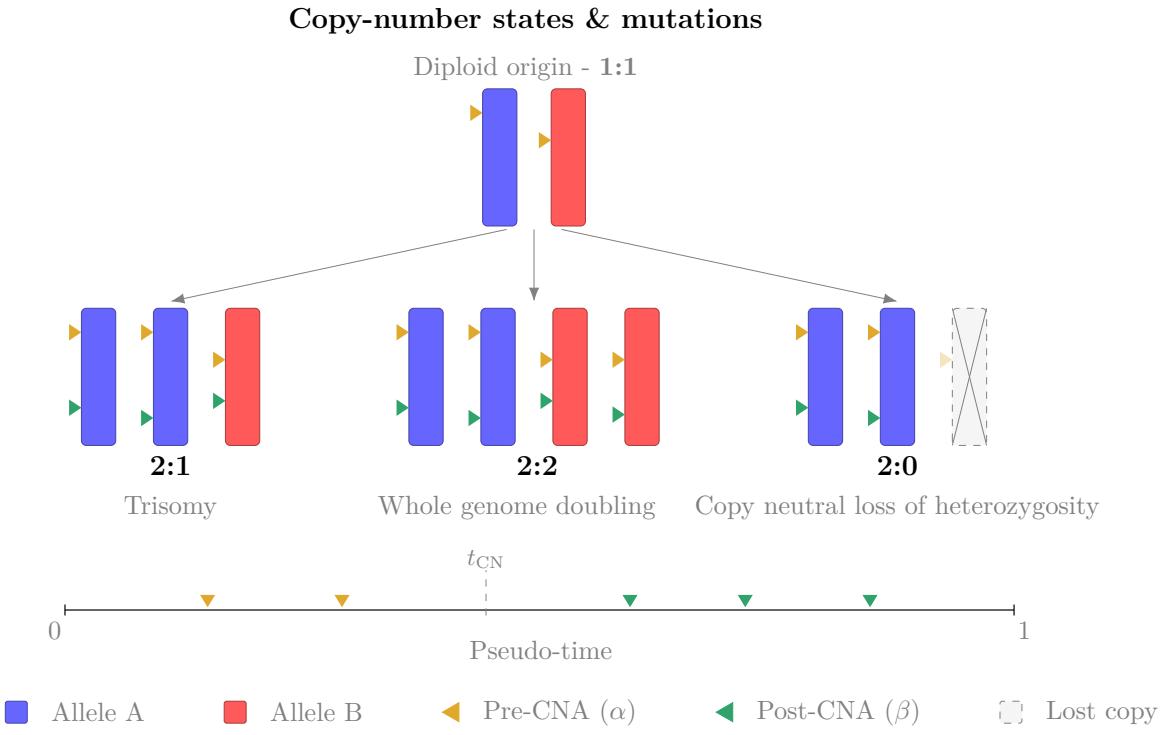



Figure 1.1: Top: copy-number states 2:1, 2:2, 2:0 from a diploid 1:1 origin. Pre-CNA mutations (gold, $\alpha$) sit at a fixed locus on every surviving copy of their allele; post-CNA mutations (green, $\beta$) are independent per copy. In 2:0 the lost allele (dashed) removes its mutation. Bottom: pseudo-time, with $\alpha$ before and $\beta$ after the CNA time $t_{\mathrm{CN}}$.

### 1.1.1 Sequencing

Sequencing measures the genetic makeup of a tumour and provides information about somatic mutations, copy-number changes, and clonality. Two sequencing strategies are relevant to this work: bulk sequencing and single-cell sequencing.

Bulk sequencing extracts DNA from the whole cell population, so the signal it returns is averaged over all the sampled cells. For each genomic site one observes the *variant allele frequency* (VAF), the fraction of sequencing reads at that site that carry the mutation (Cocomello et al. 2025). Bulk sequencing is cheap and widely used, but the averaging hides which cells carry which mutations, so reconstructing the clonal composition from VAFs alone is difficult.

Single-cell sequencing measures cells one at a time. This makes it possible to see which mutations co-occur in the same cell, to assign cells to clones, and to reconstruct the phylogenetic tree. The gain in resolution is what makes single-cell data so valuable for studying tumour evolution. The price is noise: because each cell contains very little DNA, it must be amplified, and amplification introduces allelic dropout, false positives, and uneven, sometimes insufficient, sequencing depth. As a result, not finding a mutation in a given cell does not mean the mutation is absent.

Single-cell sequencing is the basis of this work: the assignment of cells to clones and the number of mutations along each branch of the tree give us the information we need about event timings and growth rates. The limitations described above concern *accuracy*, the noise in the measured genotypes, and not the tree topology itself. A separate issue is *sampling*: because only part of the tumour is sequenced, the reconstructed phylogeny, and in particular which clone appears as the most recent common ancestor (MRCA), depends on which cells are sampled and need not match the true evolutionary history. Figure 1.2 illustrates how different samples of the same tumour can yield different reconstructed trees and a different MRCA.

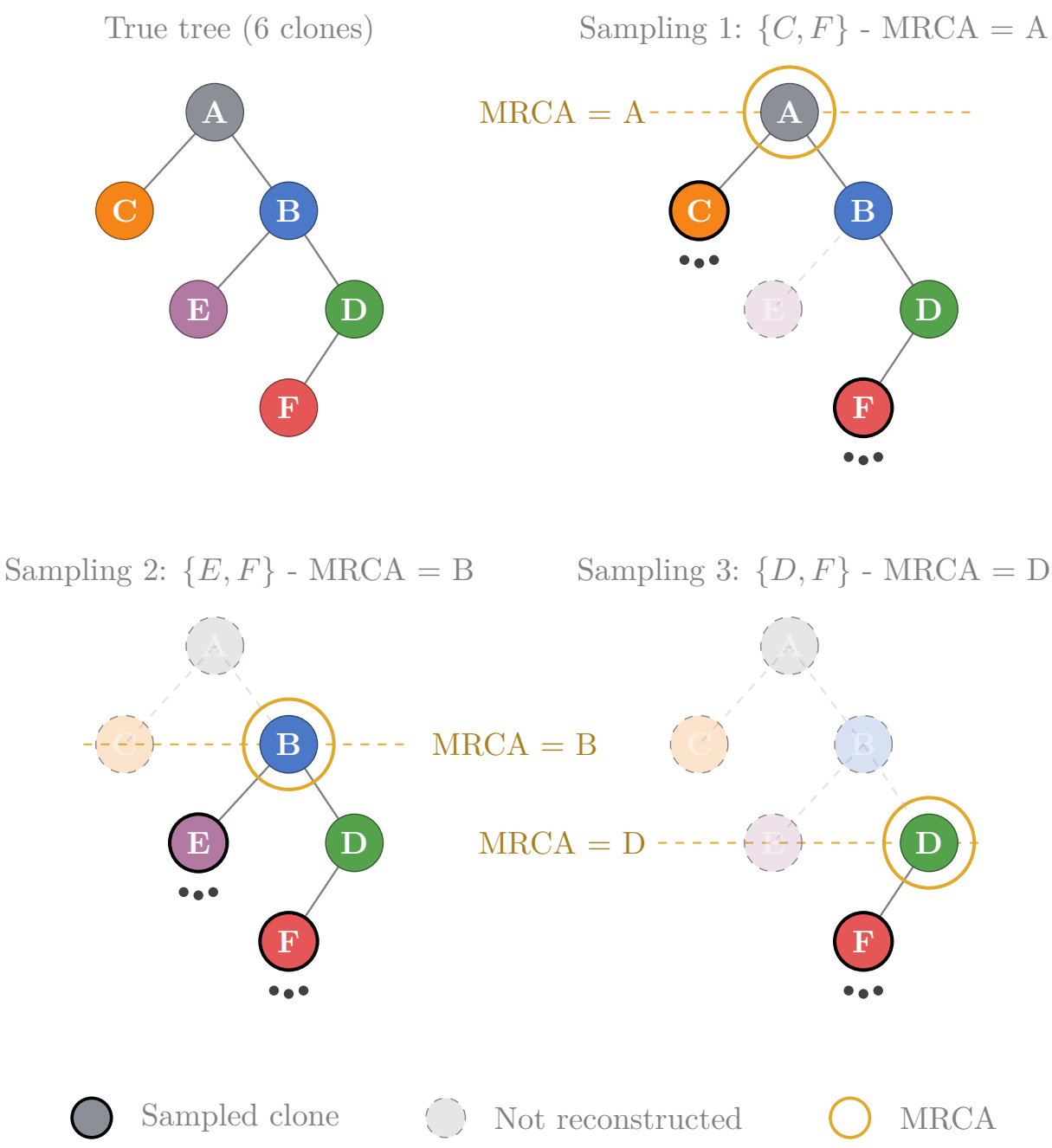



Figure 1.2: Examples of how different samplings lead to different reconstructed trees

### 1.1.2 Real Data and ProCESS

Validation of the proposed tumour evolution model on authentic single-cell data is clearly the ultimate aim because the latter encapsulates the complexity of the biological process. However, several issues make such validation problematic. Datasets of single-cell sequencing are expensive to obtain, hard to access, and many publicly available cancer data were not collected with the aim of reconstructing clonal evolution, often lacking the mutation numbers, clone assignments or a phylogenetic tree needed for our model. More importantly, there is no ground truth: even after inference, sequencing-based phylogeny cannot be considered to be accurate, and timings of the evolutionary events and the growth rates of clones are always unknown. Hence, the validation must rely on simulation.

The first step of the validation will be conducted assuming that the sequence was generated according to the process defined by the model. Inability to reproduce the generating parameter values would mean that the inference algorithm either fails in some particular cases or there is no enough information contained in the dataset.

Another more stringent validation procedure will be carried out based on the Programmable Cancer Evolution Spatial Simulator (ProCESS). As opposed to simplified simulation methods used for data generation, ProCESS produces realistic cancer histories while keeping track of all process parameters and allowing us to compare them directly with our inferred estimates. Moreover, ProCESS enables single-cell sampling from the tumour in question, allowing us to create a partially and possibly biased sample similar to sequencing data. Additionally, simulations under the same parameter settings can be repeated multiple times for evaluating stability of the inference results.

Hence, ProCESS provides a compromise between the idealistic simulation and real data as it maintains biologically relevant details, yet retains perfect control over the data-generating process and ground truth information.

### 1.1.3 The Inference Problem

The goal of this thesis can be stated in one sentence: from a single snapshot of single-cell sequencing data, reconstruct *when* the events in a tumour's history happened and *how fast* each clone grew. For a tumour made of $K$ clones, this means estimating the birth time of every clone, the time of every copy-number alteration, and the growth rate of every clone, each reported with its uncertainty rather than as a single point estimate.

We start from a specific set of inputs. The first is the reconstructed phylogenetic tree over the clones, obtained from the single-cell data by an aforementioned method. What the tree provides are the *parental relations*: for each clone it tells us the parent clone from which it descends. These relations are the backbone of the problem, because they fix the *order* of the events even when their exact times are unknown: a clone is always younger than its parent, and a copy-number alteration on a branch must have occurred after the parent was born and before the child split off. Writing $t_i$ for the birth time of clone $i$ on a scale where the tumour starts at 0 and sequencing happens at 1, the tree thus imposes

$$0 < t_{\mathrm{pa}(i)} < t_i < 1. \tag{1.1}$$

We do not treat the tree as exact ground truth; we take its structure, who descends from whom, as the scaffold on which the timing is inferred.

On top of this scaffold, three kinds of measurement carry the information we exploit: the number of somatic mutations along each branch, which acts as a clock for how long the branch lasted; for the regions affected by a copy-number alteration, the split of those mutations into the ones that arose before and after the event, which locates the alteration within its branch; and the number of cells assigned to each clone, which reflects how much the clone expanded before sequencing.

These signals do not pin down the timings on their own. The number of mutations and the size of a clone both depend on the combination of how fast it grew and how long it lived, so a clone that grew quickly for a short time can look much like one that grew slowly for longer. Separating the timing of an event from the growth of the clone is therefore

the central difficulty of the problem. The model developed in Chapter 2 addresses it by combining these sources of evidence in a single Bayesian framework and by sharing growth information between related clones along the tree, so that timings and growth rates are recovered together, with calibrated uncertainty.

## 1.2 Statistical Framework

### 1.2.1 Brief introduction to bayesian inference

Bayesian inference is a consistent mathematical formalism for learning about the underlying quantities from observations. While the frequentist framework treats parameters as fixed but unknown constants, the Bayesian one regards parameters as random variables and makes use of Bayes' rule to update a prior distribution in light of observed data.

Bayes' theorem, the foundation of Bayesian inference, relates the posterior probability distribution $p(\theta \mid \mathbf{y})$, representing our beliefs about unknown parameters $\theta$ given data $\mathbf{y}$, the likelihood $p(\mathbf{y} \mid \theta)$, reflecting our belief about how likely the observed data were generated under different $\theta$ values, and the prior $p(\theta)$:

$$p(\theta \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \theta)\, p(\theta)}{p(\mathbf{y})}, \tag{1.2}$$

where $p(\mathbf{y}) = \int p(\mathbf{y} \mid \theta)\, p(\theta)\, \mathrm{d}\theta$ is called the marginal likelihood (also known as evidence or partition function) and serves as a normalising factor. Since the marginal likelihood does not depend on the value of the unknown parameters, the posterior distribution may equivalently be written as

$$p(\theta \mid \mathbf{y}) \propto p(\mathbf{y} \mid \theta)\, p(\theta). \tag{1.3}$$

The above equation captures the essence of Bayesian inference, namely that the posterior is proportional to the product of the likelihood and the prior.

An important feature of the Bayesian framework is that it gives rise to different forms of priors. If there is sufficient prior knowledge about the parameters, one can use an informative prior and concentrate prior probability mass around plausible parameter values. Weakly

informative priors act as very mild regularisers with little effect on the posterior distribution. On the other hand, in many problems, data alone is insufficient to recover the parameter values; that is, there is no unique solution to the problem. In such problems, the prior acts as an additional source of information, ensuring that the resulting posterior is well-regularised. The latter scenario is encountered in the current thesis because mutation counts and clone abundances only constrain the product $w_i \, \Delta t_i$, not its two factors separately: a clone that grew quickly for a short time looks the same as one that grew slowly for longer. It is precisely because the data cannot separate $w_i$ from $\Delta t_i$ that we introduce a prior, which supplies the information the likelihood is missing along this direction and yields a well-defined posterior.

One generalisation of the above framework is the hierarchical model. In this setup, the parameters themselves are treated as random variables whose distribution depends on some additional hyperparameters. More formally, let $\theta_1, \ldots, \theta_K$ be parameters whose values are distributed according to

$$\theta_i \mid \phi \sim p(\theta_i \mid \phi), \quad i \in \{1, \ldots, K\}. \tag{1.4}$$

Here, $\phi$ denotes hyperparameters, and we impose a hyperprior $p(\phi)$ on $\phi$. It follows that the joint posterior distribution is given by

$$p(\theta_1, \ldots, \theta_K, \phi \mid \mathbf{y}) \propto p(\mathbf{y} \mid \theta_1, \ldots, \theta_K) \prod_{i=1}^{K} p(\theta_i \mid \phi) \, p(\phi). \tag{1.5}$$

One of the key advantages of the hierarchical approach is partial pooling, where information from one unit is pooled into the rest of the units, making it possible for poorly informed units to borrow strength from better-informed ones. In the model proposed in this thesis, a hierarchical prior is imposed on clone-specific growth rates based on a random walk on the log scale along phylogenetic branches.

In practice, the posterior distribution $p(\theta \mid \mathbf{y})$ is rarely obtainable analytically, so we need to rely on some approximation method. The leading methodology used in applied Bayesian inference is Markov chain Monte Carlo (MCMC), which builds a Markov chain whose limiting distribution coincides with the desired posterior. Posterior quantities are estimated from post-burn-in Markov chain samples. An efficient MCMC scheme is Hamiltonian Monte

Carlo (HMC) and its adaptive version, the No-U-Turn Sampler (NUTS). Both use gradient information from the log-posterior to generate distant jumps, ensuring that the chain mixes quickly even in high-dimensional or highly correlated spaces. The models proposed in this thesis are coded in Stan, a probabilistic programming language that automatically computes gradients and implements NUTS as its default sampler.

### 1.2.2 Hamiltonian Monte Carlo

Hamiltonian Monte Carlo (HMC) represents one approach to Markov chain Monte Carlo sampling inspired by classical mechanics. This involves defining a new momentum variable $\rho$, which must have the same dimensions as $\theta$, according to

$$H(\theta, \rho) = -\log p(\theta \mid \mathbf{y}) + \frac{1}{2}\,\rho^\top M^{-1}\rho, \tag{1.6}$$

with $M$ representing the mass matrix, normally set to the identity matrix. The first term can be viewed as potential energy and the second as kinetic energy. Sampling proceeds through Hamiltonian dynamics generated by numerically integrating the equations of motion

$$\frac{\mathrm{d}\theta}{\mathrm{d}s} = M^{-1}\rho, \tag{1.7}$$

$$\frac{\mathrm{d}\rho}{\mathrm{d}s} = \nabla_\theta \log p(\theta \mid \mathbf{y}), \tag{1.8}$$

using a leapfrog integrator with integration step size $\varepsilon$ and $L$ leapfrog steps. A Metropolis correction is applied to the result to account for any errors in the integration. In essence, the leapfrog algorithm makes use of gradients of the log-posterior to generate distant proposals, which are rarely rejected, resulting in chains exploring the posterior distribution better than random walk samplers.

One of the major issues with Hamiltonian Monte Carlo is setting the hyperparameters $\varepsilon$ and $L$ appropriately. The No-U-Turn Sampler (NUTS) eliminates this problem by automatically computing the value of $L$, increasing it until a reversal in the leapfrog trajectory occurs, explaining its name. Step size tuning is performed automatically as part of a burn-in period. NUTS is the default sampler used by Stan, and the gradients $\nabla_\theta \log p(\theta \mid \mathbf{y})$ are automatically calculated using automatic differentiation.

# 2. Model

## 2.1 Basic Model: Timing Reconstruction with Known Growth Rates

The first model assumes that the growth rates $w_1, \ldots, w_K$ are fixed and known, and makes inference only on clone birth event timings and copy-number alteration event timings. This simple framework serves as a stepping stone to develop a more general model described in the next section, providing an opportunity to verify whether the inference procedure works correctly under circumstances when the dynamics of the expansions do not induce any uncertainty.

**Definition 2.1** (Pseudotime)**.** Time is measured on the normalised scale $t \in [0, 1]$, the *pseudotime*, with $t = 0$ the origin of the tumour and $t = 1$ the moment of sequencing. All event times in this chapter are expressed in pseudotime.

Unknown parameters include clone birth event timings and copy-number alteration event timings. For each node $i$ of the phylogenetic tree $\mathcal{T}$, where $i \in \{1, \ldots, K\}$, let $t_i \in (0, 1)$ be the time of birth of clone $i$, with time normalised such that $t_0 = 0$ marks the origin of the tumour and $T_f = 1$ marks the moment of sequencing. Further, for each copy-number alteration event $j$ corresponding to node $i$, let $t_{CN,ij}$ be its timing. The phylogenetic tree specifies ordering constraints

$$0 < t_{\mathrm{pa}(i)} < t_i < 1, \quad i = 2, \ldots, K, \tag{2.1}$$

where $\mathrm{pa}(i)$ stands for the parent of node $i$ in $\mathcal{T}$, and each copy-number alteration occurs in the branch of the node to which it is linked,

$$t_{\mathrm{pa}(i)} < t_{CN,ij} < t_i. \tag{2.2}$$

To satisfy the constraints and at the same time allow a representation that is amenable to sampling via gradients, all event times are parametrised using auxiliary uniformly distributed

variables $u \sim \text{Uniform}(0, 1)$. Clone birth times are represented as $t_1 = u_1$ for the root node ($i = 1$, which has no parent), and

$$t_i = t_{\text{pa}(i)} + u_i \cdot \Big(1 - t_{\text{pa}(i)}\Big) \tag{2.3}$$

for non-root nodes. Copy-number alteration event times are parametrised similarly: $t_{CN,1j} = u_{1j} \cdot t_1$ for copy-number alteration events on the root branch, and

$$t_{CN,ij} = t_{\text{pa}(i)} + u_{ij} \cdot \Big(t_i - t_{\text{pa}(i)}\Big) \tag{2.4}$$

for other branches. Since all auxiliary variables are uniformly distributed on $(0, 1)$, these mappings imply a uniform prior over the event times:

$$t_i \sim \text{Uniform}\Big(t_{\text{pa}(i)},\, T_f\Big), \qquad t_{CN,ij} \sim \text{Uniform}\Big(t_{\text{pa}(i)},\, t_i\Big). \tag{2.5}$$

There are two independent components of the likelihood, one related to mutation counts and another related to clone abundances.

First, we define the likelihood based on mutations. Mutations accumulate independently along each branch as a stochastic process, with an expected number that depends on the mutation rate $\mu$, the growth rate $w_i$, the length of the genomic region considered, and the duration of the branch. The relevant distinction is not whether a branch carries a copy-number alteration, but which *segments* of the genome do: a branch generally contains both segments that remain diploid and, possibly, segments altered by a copy-number event, and the two contribute to the likelihood in different ways.

For the segments that remain diploid along the branch of clone $i$, whose total length we denote by $l_{\text{tot},i}$, the mutation count $m_i$ follows a Poisson distribution,

$$m_i \sim \text{Poisson}(r_i), \qquad r_i = \mu \cdot w_i \cdot l_{\text{tot},i} \cdot (t_i - t_{\text{pa}(i)}). \tag{2.6}$$

See Appendix B.1 for the explanation of why this likelihood emerges naturally from the solution of the system of equations relating mutation counts to unknown event times, and for the corresponding deterministic formulation.

On a genomic segment altered by a copy-number event $j$ at time $t_{CN,ij}$, the mutations falling in that segment carry information about the timing of the alteration. They are split into pre-alteration mutations $\alpha_{ij}$ and post-alteration mutations $\beta_{ij}$, with their respective Poisson likelihoods,

$$\alpha_{ij} \sim \text{Poisson}(r^{\alpha}_{ij}), \qquad r^{\alpha}_{ij} = \mu \cdot w_i \cdot l_{CN\text{tot},ij} \cdot c^{\alpha}_{ij} \cdot (t_{CN,ij} - t_{\text{pa}(i)}), \tag{2.7}$$

$$\beta_{ij} \sim \text{Poisson}(r^{\beta}_{ij}), \qquad r^{\beta}_{ij} = \mu \cdot w_i \cdot l_{CN\text{tot},ij} \cdot c^{\beta}_{ij} \cdot (t_i - t_{CN,ij}), \tag{2.8}$$

where the scaling factors $c^{\alpha}_{ij}$ and $c^{\beta}_{ij}$ correspond to the number of allelic copies on which new mutations may accumulate before and after the alteration respectively. These scalings differ for different types of copy-number alteration:

$$c^{\alpha}_{ij} = \begin{cases} 2 & \text{if } CN_{ij} = 2{:}2, \\ 1 & \text{if } CN_{ij} = 2{:}0, \\ 2 & \text{if } CN_{ij} = 2{:}1, \end{cases} \qquad c^{\beta}_{ij} = \begin{cases} 4 & \text{if } CN_{ij} = 2{:}2, \\ 2 & \text{if } CN_{ij} = 2{:}0, \\ 3 & \text{if } CN_{ij} = 2{:}1. \end{cases} \tag{2.9}$$

Finally, we define the likelihood of the observed clone abundance $N_i$. Since the descendant clones occupy part of this mass, the cells attributable to clone $i$ itself are obtained by subtracting the contributions of its child clones, yielding the terminal mass $M_i$; normalising the terminal masses across clones gives the expected proportion $p_i$, so that the observed cell count $N_i$ follows a Negative Binomial distribution,

$$\begin{aligned} M_i &= \exp\Big[w_i \cdot (T_f - t_i)\Big] - \sum_{j:\,\text{pa}(j)=i} \exp\Big[w_i \cdot (T_f - t_j)\Big], \qquad p_i = \frac{M_i}{\sum_k M_k}, \\ N_i &\sim \text{NegBin}\Big(N_{\text{tot}} \cdot p_i,\, \phi\Big), \qquad N_{\text{tot}} = \sum_k N_k, \end{aligned} \tag{2.10}$$

where the dispersion parameter $\phi$ controls how much the observed counts may deviate from their expected proportions. The Negative Binomial generalises the Poisson by adding exactly this dispersion, and a single $\phi$ lets us collect the different sources of extra variability in one place: the randomness of which cells end up in the sample, the stochastic fluctuations of clone proliferation, and the noise introduced by the sequencing process itself, such as the misassignment of cells to clones caused by allelic dropout or false positives. We assume the

same value of $\phi$ for all clones, so that the number of parameters does not increase with the number of clones. The full posterior probability density of the event times $\boldsymbol{t}$ and copy-number alteration times $\boldsymbol{t}_{CN}$ can be written in terms of the prior and the two likelihood components as

$$p(\boldsymbol{t}, \boldsymbol{t}_{CN} \mid \boldsymbol{m}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{N}, \mathcal{T}) \propto \left[ \prod_i p(m_i \mid t_i, t_{\mathrm{pa}(i)}) \prod_{i,j} p(\alpha_{ij}, \beta_{ij} \mid t_{CN,ij}, t_i, t_{\mathrm{pa}(i)}) \right] \cdot \left[ \prod_i p(N_i \mid t_i) \right] \cdot \left[ \prod_i p(t_i) \prod_{i,j} p(t_{CN,ij}) \right], \tag{2.11}$$

where the last term refers to the uniform priors over the event times. Since it is not possible to sample from this posterior in closed form, we approximate it using the No-U-Turn Sampler as implemented in Stan.

Putting the components together, the basic model has a simple two-layer structure: the latent event times $\boldsymbol{t}$ and $\boldsymbol{t}_{CN}$, with the uniform priors above, generate conditionally independently the mutation counts $m_i$, $\alpha_{ij}$, $\beta_{ij}$ through the Poisson likelihoods and the clone abundances $N_i$ through the Negative-Binomial likelihood, while the growth rates $w_i$ are held fixed.

### 2.1.1 Probabilistic graphical model

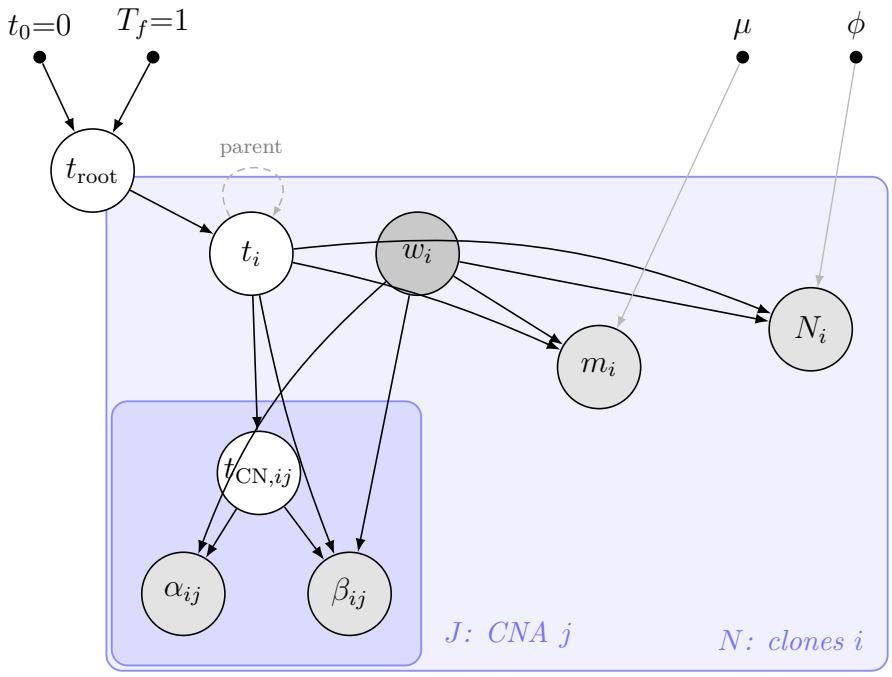



Figure 2.1: Probabilistic graphical model of the base model

### 2.1.2 Hierarchical formulation

$$
\begin{aligned}
u_i,\, u_{ij} &\sim \text{Uniform}(0,1) \\
t_1 &= u_1 \\
t_i &= t_{\text{pa}(i)} + u_i\Big(1 - t_{\text{pa}(i)}\Big) \\
t_{\text{CN},ij} &= t_{\text{pa}(i)} + u_{ij}\Big(t_i - t_{\text{pa}(i)}\Big) \\
m_i &\sim \text{Poisson}(r_i) \\
r_i &= \mu\, w_i\, l_{\text{tot},i}\Big(t_i - t_{\text{pa}(i)}\Big)
\end{aligned}
\qquad
\begin{aligned}
\alpha_{ij} &\sim \text{Poisson}(r^{\alpha}_{ij}) \\
r^{\alpha}_{ij} &= \mu\, w_i\, l_{\text{CN},ij}\, c^{\alpha}_{ij}\Big(t_{\text{CN},ij} - t_{\text{pa}(i)}\Big) \\
\beta_{ij} &\sim \text{Poisson}(r^{\beta}_{ij}) \\
r^{\beta}_{ij} &= \mu\, w_i\, l_{\text{CN},ij}\, c^{\beta}_{ij}\Big(t_i - t_{\text{CN},ij}\Big) \\
M_i &= e^{\,w_i(T_f - t_i)} - \sum_{j:\text{pa}(j)=i} e^{\,w_i(T_f - t_j)} \\
p_i &= M_i \Big/ \textstyle\sum_k M_k \\
N_i &\sim \text{NegBin}(N_{\text{tot}}\, p_i, \phi)
\end{aligned}
$$

## 2.2 Extended Model: Joint Reconstruction of Times and Growth Rates

Model two generalises the previous one by removing the assumption that the growth rates associated with clones are given as input data. In this model, the growth rates $w_1, \ldots, w_K$ are considered latent variables, together with the times at which the events occurred. Again, the likelihood and the parametrisation of event times are identical to what was explained in the previous section, the only difference being the hierarchical structure assumed over the growth rates.

The main idea underlying this prior comes from the observation that the growth rates for related clones are similar, since these clones have been evolving together over a large amount of time. This idea is incorporated into the hierarchy via a log-scale random walk over the branches of the tree $\mathcal{T}$. Specifically, we define a global mean $\bar{w} > 0$, and then compute the growth rate of each node $i$ as

$$\log w_i = \log w_{\mathrm{pa}(i)} + \sigma_{\log w} \cdot z_i, \qquad z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{2.12}$$

where $\sigma_{\log w} \geq 0$ controls the variance of the variation in growth rates between parent and child clones. If $\sigma_{\log w}$ is close to zero, related clones are forced to have very similar growth rates; if it is large, their rates may differ considerably. In the implementation the per-branch increments $\delta_i = \log w_i - \log w_{\mathrm{pa}(i)}$ are sampled directly with $\delta_i \sim \mathcal{N}(0, \sigma_{\log w})$ (a centred parametrisation), which proved sufficient for stable sampling with the No-U-Turn Sampler.

The prior distributions for this hierarchical model are defined as follows. First, the global mean growth rate is assigned a log-normal prior,

$$\log \bar{w} \sim \mathcal{N}(\log \bar{w}_{\mathrm{prior}},\, \sigma_{\bar{w}})\,, \tag{2.13}$$

with $\bar{w}_{\mathrm{prior}}$ and $\sigma_{\bar{w}}$ being hyperparameters passed as input. The centre $\bar{w}_{\mathrm{prior}}$ is chosen to match the growth rates reported for real tumours rather than set arbitrarily. Writing the specific growth rate as $\mathrm{SGR} = \ln 2 / T_d$, with $T_d$ the volume doubling time (Mehrara et al. 2007), the doubling times observed for solid tumours, from a few months up to about a year,

correspond to rates of the order of 0.05 to 0.2 per month. We therefore centre $\bar{w}_{\text{prior}}$ near the lower end of this interval, a doubling time of roughly one year, which keeps the prior on a biologically plausible scale while the spread $\sigma_{\bar{w}}$ keeps it weakly informative. The step size of the walk is given an exponential prior,

$$\sigma_{\log w} \sim \text{Exponential}(\lambda), \tag{2.14}$$

where $\lambda$ controls the strength of shrinkage towards the global mean rate. A large value of $\lambda$ implies that $\sigma_{\log w}$ will be small, making related growth rates more similar. Finally, the standardised increments are drawn from the standard normal distribution,

$$z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \quad i = 2, \ldots, K. \tag{2.15}$$

The full posterior distribution is given by

$$\begin{aligned} p(\boldsymbol{t}, \boldsymbol{t}_{CN}, \bar{w}, \sigma_{\log w}, \boldsymbol{z} \mid \boldsymbol{m}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{N}, \mathcal{T}) \propto & \left[\prod_i p(m_i \mid t_i, t_{\text{pa}(i)}, w_i) \prod_{i,j} p(\alpha_{ij}, \beta_{ij} \mid t_{CN,ij}, t_i, t_{\text{pa}(i)}, w_i)\right] \\ & \times \left[\prod_i p(N_i \mid t_i, w_i)\right] \\ & \times \left[\prod_i p(t_i) \prod_{i,j} p(t_{CN,ij})\right] \\ & \times \; p(\bar{w})\, p(\sigma_{\log w}) \prod_{i=2}^{K} p(z_i), \end{aligned} \tag{2.16}$$

where $\sigma_{\log w} \geq 0$ controls the variance of the variation in growth rates between parent and child clones. If $\sigma_{\log w}$ is close to zero, related clones are forced to have very similar growth rates; if it is large, their rates may differ considerably. See Appendix B.2 for a derivation of the marginal distribution of $w_i$ implied by this hierarchical structure.

The extended model adds a further layer on top of this structure. At the top, the global parameters $\bar{w}$ and $\sigma_{\log w}$ are drawn from their priors; the standardised increments $z_i$ then generate the clone growth rates $w_i$ through the log-scale random walk; finally, the growth rates together with the event times generate the observed mutation counts and the clone abundances. The event times and the growth rates are therefore the latent nodes shared be-

tween the two likelihood branches, and the joint density factorises exactly as in the posterior above.

### 2.2.1 Probabilistic graphical models

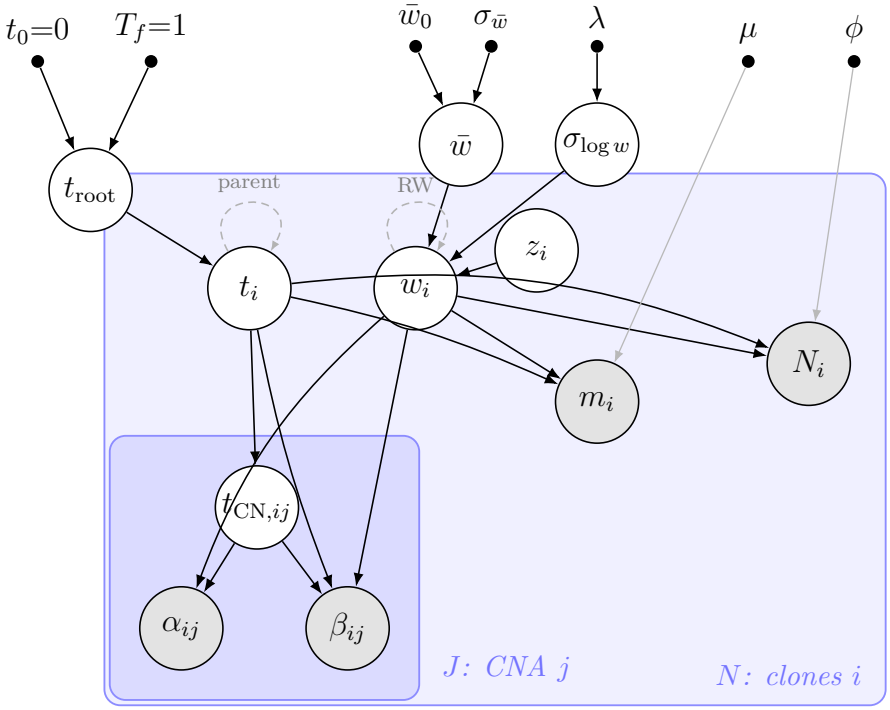



Figure 2.2: Probabilistic graphical model of the complete model

### 2.2.2 Hierarchical formulation

$$
\begin{aligned}
u_i,\, u_{ij} &\sim \text{Uniform}(0,1) \\
\bar{w} &\sim \text{LogNormal}(\log \bar{w}_0, \sigma_{\bar{w}}) \\
\sigma_{\log w} &\sim \text{Exponential}(\lambda) \\
z_i &\sim \mathcal{N}(0,1) \\
t_1 &= u_1 \\
t_i &= t_{\text{pa}(i)} + u_i(1 - t_{\text{pa}(i)}) \\
t_{\text{CN},ij} &= t_{\text{pa}(i)} + u_{ij}(t_i - t_{\text{pa}(i)}) \\
\log w_1 &= \log \bar{w} \\
\log w_i &= \log w_{\text{pa}(i)} + \sigma_{\log w} z_i
\end{aligned}
$$

$$
\begin{aligned}
m_i &\sim \text{Poisson}(r_i) \\
r_i &= \mu w_i l_{\text{tot},i}(t_i - t_{\text{pa}(i)}) \\
\alpha_{ij} &\sim \text{Poisson}(r^{\alpha}_{ij}) \\
r^{\alpha}_{ij} &= \mu w_i l_{\text{CN},ij} c^{\alpha}_{ij}(t_{\text{CN},ij} - t_{\text{pa}(i)}) \\
\beta_{ij} &\sim \text{Poisson}(r^{\beta}_{ij}) \\
r^{\beta}_{ij} &= \mu w_i l_{\text{CN},ij} c^{\beta}_{ij}(t_i - t_{\text{CN},ij}) \\
M_i &= e^{w_i(T_f - t_i)} - \sum_{j:\text{pa}(j)=i} e^{w_i(T_f - t_j)} \\
p_i &= M_i / \textstyle\sum_k M_k \\
N_i &\sim \text{NegBin}(N_{\text{tot}} p_i, \phi)
\end{aligned}
$$

# 3. Simulation Study and Model Validation

Validating the model is done in two stages, in order of increasing difficulty. We first feed the model data produced by exactly the process it assumes, the *ideal* (well-specified) data-generating process, and ask whether the inference returns the truth; this is the subject of the first part of this chapter. Only afterwards do we expose the model to synthetic data from ProCESS, a cancer-evolution simulator.

## 3.1 Simulation under the Ideal Data-Generating Process and Robustness Check

Every experiment in this section draws its data from a single generative process that mirrors the random-walk model of Section 2.2 exactly: clone growth rates follow a log-scale random walk along the tree, event times are placed by stick-breaking, somatic mutations are Poisson distributed with intensity proportional to the clone-specific molecular clock, and clone abundances are drawn from a negative-binomial distribution on the simplex of terminal masses. Because the data come from the model itself, any failure to recover the truth must be blamed on the inference, on identifiability, on the sampler, or simply on the limited information carried by a single snapshot, rather than on a mismatch between model and data. We proceed in three steps: parameter recovery, calibration in the stronger sense of simulation-based calibration, and a set of sensitivity checks on the prior choices and the amount of data.

### 3.1.1 Parameter recovery under the ideal data-generating process

The recovery experiment follows the standard template used in Bayesian workflows (Schad, Betancourt, and Vasishth 2021). For each replicate we draw a set of true parameters from the priors, simulate a data set conditional on them, fit the model, and compare the posterior

with the values that actually generated the data; repeating this over many replicates sweeps the parameter space rather than testing a single lucky configuration.

**Ideal DGP** Nnodes=4 | Nevents=6 | Nequations=8

**Topology** (branching, 4 clones, 0 hidden)

$\texttt{Clone_A} \longrightarrow \texttt{Clone_B} \longrightarrow \left\{ \texttt{Clone_C}, \texttt{Clone_D} \right\}$

| | |
|---|---|
| **Clones (all sampled)** | Clone_A, Clone_B, Clone_C, Clone_D |
| **Stan parents** | (0, 1, 2, 2) |
| **number_of_events** | (1, 1, 2, 2) |
| **CNAs** | gain on Clone_C and on Clone_D |
| **Segment length** | diploid/MRCA 10, CNA 5 (arbitrary units) |
| **Growth** | per-clone, log–random-walk along the tree |
| **Growth priors** | $\bar{w} \sim \text{LogN}(\log 0.06,\, 0.35)$, $\sigma_{\log w} \sim \text{Exp}(1)$, $\omega \sim \text{LogN}(\log 1.25,\, 0.5)$ |
| **Event times** | nested-uniform stick-breaking, prior on $[0, 1]$ |
| **Mutations** | $m \sim \text{Poisson}(\nu\, \mu\, \Delta t\, \ell\, c)$, $\mu = 3$ |
| **Abundances** | $N_i \sim \text{NegBin}(N\, p_i,\, \varphi)$, $\varphi = 100$ |
| **Equal rates** | none (distinct by construction) |

The replicate shown below is one such draw from the process summarised in the card above. Its four clones grow at rates $w_A = 0.081$, $w_B = 0.381$, $w_C = 0.133$, $w_D = 0.701$ and are born at pseudotimes 0.21, 0.80, 0.97, 0.88 (clones $A$–$D$; real times 14.1, 25.9, 29.4, 27.7 on the $[t_0, T_f] = [10, 30]$ axis), with the two copy-number events at $t = 0.91$ and $t = 0.85$. The simulated sample holds 2526 cells, split unevenly across the clones (658, 301, 333 and 1234 for $A$–$D$): even the ideal process yields skewed clone proportions.

Figures 3.1 and 3.2 show, for this representative replicate, the posterior of the event times and of the clone growth rates against the values that generated the data. The event-time posteriors concentrate tightly around the truth, with the true value sitting inside the shaded

95% credible interval for every event. The growth-rate posteriors are visibly wider and more skewed, especially for the faster-growing clones: the model still covers the truth, but it reports this through broad intervals rather than sharp point estimates, the first sign of the single-snapshot identifiability limit discussed below.

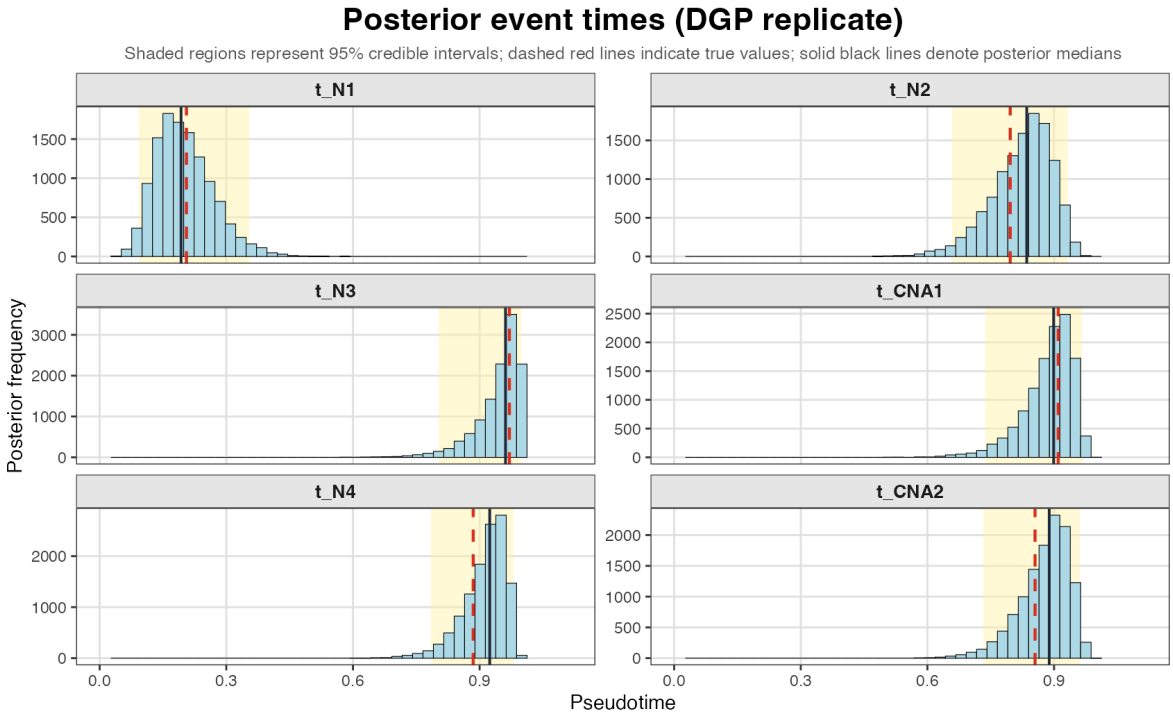



Figure 3.1: Posterior event times for a representative replicate simulated under the ideal data-generating process. Shaded regions are 95% credible intervals, the dashed red lines mark the true values, and the solid lines the posterior medians.

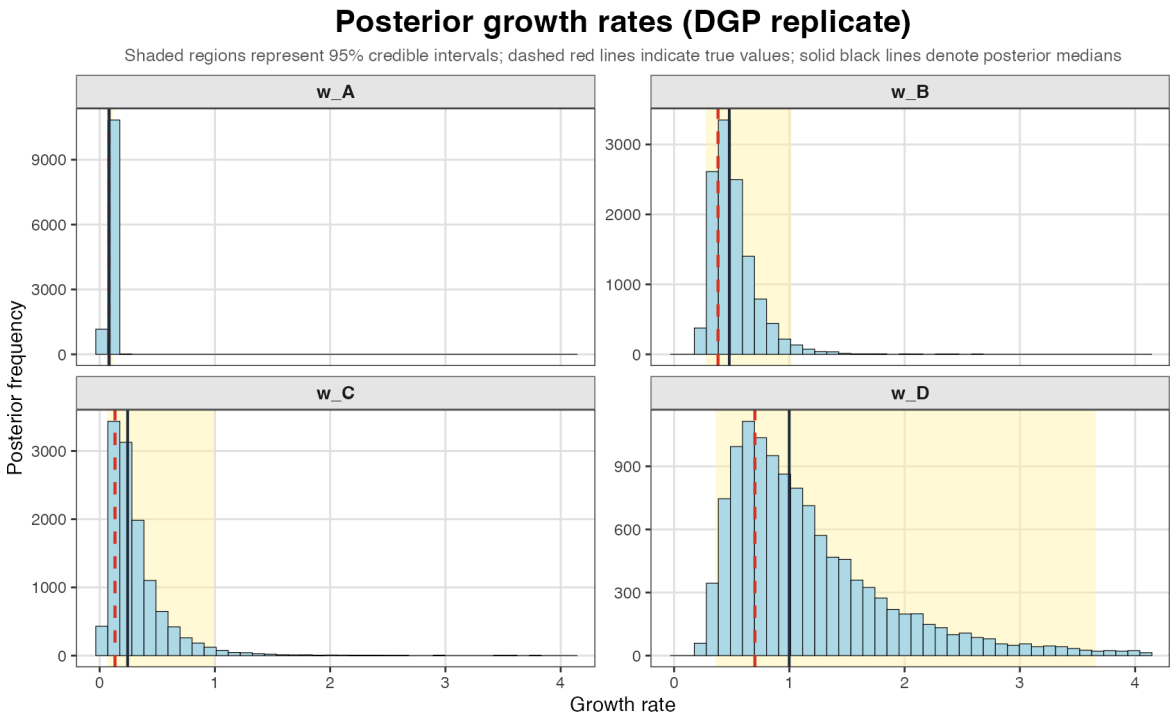



Figure 3.2: Posterior clone growth rates for the same replicate; conventions as in Figure 3.1. The intervals are wider than for the event times, reflecting the weaker identifiability of per-clone growth from a single snapshot.

A third view completes the picture. The mutation counts are the data the model conditions on, so their posterior-predictive distribution (the `mutations_rep` replicates of Appendix A.2) measures whether the fit can reproduce what it was given. Figure 3.3 shows, for the same replicate, the predictive distribution of each of the eight mutation equations with the observed count overlaid. Every observed value lies inside its 95% posterior-predictive interval, and the predictive medians track the observations closely across counts that range from a single mutation to several dozen: the likelihood reproduces the data it was conditioned on, as it should under the ideal process.

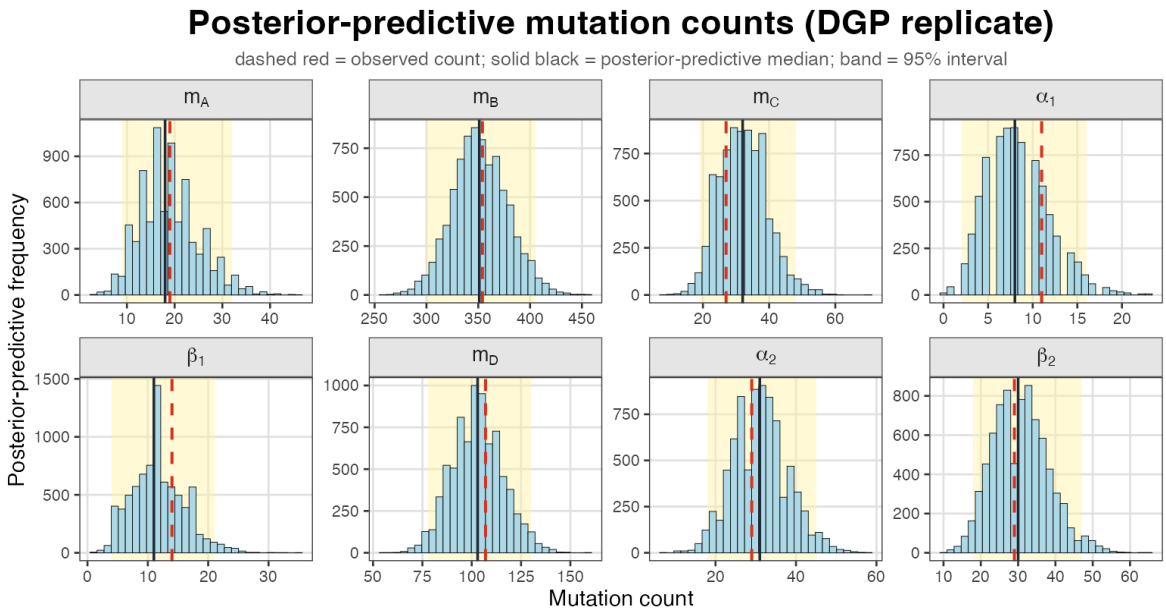



Figure 3.3: Posterior-predictive distribution of the mutation counts for the same DGP replicate. Each panel is one mutation equation; the dashed red line is the observed count, the solid line the predictive median, and the shaded band the 95% posterior-predictive interval. All eight observed counts fall inside their intervals.

For this replicate the three checks agree: event times and growth rates fall within their credible intervals, and the mutation counts are reproduced. The model behaves as it should when the data come from its own assumptions.

This pattern holds in aggregate over the 95 of 100 replicates that converged. The event times are recovered accurately and their credible intervals are well calibrated: the mean absolute error is about 6% of the $[t_0, T_f]$ window, and the true time lies inside the 95% credible interval in 97% of cases, essentially the nominal rate. The clone growth rates are a different case. Their point estimates carry a sizeable relative error, around 23%, yet their intervals stay calibrated, covering the truth in about 97% of replicates: faced with a quantity it cannot pin down, the model does not become over-confident but widens the intervals instead. This is the single-snapshot identifiability limit: one time point constrains the *product* of a clone's growth rate and its elapsed time much better than either factor on its own, so the model can stay calibrated even where it cannot identify the point value. The simulation-based calibration of the next section reaches the same verdict more formally;

exposing this floor here, under ideal conditions, also sets a realistic expectation against which the ProCESS results later in the chapter should be judged.

### 3.1.2 Simulation-based calibration

Recovering the truth on average is reassuring, but it is not the same thing as the posterior being *calibrated*. Calibration is a statement about frequencies: if the model is correct, a 95% credible interval should contain the true value in 95% of repeated data sets. A single fit cannot speak to this, since its interval either covers the truth or it does not; calibration can only be judged by simulating many data sets and aggregating.

Simulation-based calibration (SBC) turns this idea into a concrete diagnostic (Talts et al. 2018), building on the posterior-quantile validation of Cook, Gelman, and Rubin (2006) and later refined by Modrák et al. (2023). The argument rests on a simple invariance. If a parameter is drawn from the prior and a data set is then simulated from it, the true parameter is, a priori, nothing more than one additional draw from the posterior. Its *rank* among the posterior draws is therefore uniformly distributed over its possible values.

Formally, let $\theta$ collect all model parameters (the event times, the clone growth rates, and the global scalars $\bar{w}$, $\sigma_{\log w}$ and $\omega$) and let $y$ denote the observed counts. Write the prior as $\pi(\theta)$, the likelihood as $\pi(y \mid \theta)$ and the posterior as $\pi(\theta \mid y)$. A central object here is the *prior predictive* distribution, the distribution of the data sets the model produces before any observation is seen: one first draws a parameter from the prior, $\tilde{\theta} \sim \pi(\theta)$, and then a data set from the likelihood, $y \sim \pi(y \mid \tilde{\theta})$, so that $\pi(y) = \int \pi(y \mid \tilde{\theta})\, \pi(\tilde{\theta})\, \mathrm{d}\tilde{\theta}$. The method then rests on a single exact identity: averaging the *posterior* over the prior predictive returns the prior,

$$\pi(\theta) \;=\; \iint \underbrace{\pi(\theta \mid y)}_{\text{posterior}}\; \underbrace{\pi(y \mid \tilde{\theta})\, \pi(\tilde{\theta})}_{\text{prior predictive}}\, \mathrm{d}y\, \mathrm{d}\tilde{\theta}. \tag{3.1}$$

In words: if we generate a data set from the prior predictive and then compute the posterior it induces, the parameters we recover are distributed exactly as the prior. The true value $\tilde{\theta}$ that generated the data is therefore statistically indistinguishable from an ordinary draw of that posterior. A single SBC iteration draws a ground truth from the prior, simulates a

data set from it, and fits the model,

$$\tilde{\theta} \sim \pi(\theta), \qquad \tilde{y} \sim \pi(y \mid \tilde{\theta}), \qquad \theta_1, \ldots, \theta_L \sim \pi(\theta \mid \tilde{y}), \tag{3.2}$$

and, for any one-dimensional summary $f$ of the parameters (here simply each parameter in turn), records the *rank* of the true value among the $L$ posterior draws. The rank is just the count of how many posterior draws fall below the true value: a whole number between 0, when the truth is smaller than every draw, and $L$, when it is larger than all of them,

$$r \;=\; \sum_{\ell=1}^{L} \mathbf{1}\Big[\, f(\theta_\ell) < f(\tilde{\theta}) \,\Big] \;\in\; \{0, 1, \ldots, L\}, \tag{3.3}$$

where $\mathbf{1}[\cdot]$ is the indicator that equals 1 when its condition holds and 0 otherwise. Whenever the posterior is computed exactly, this rank is uniformly distributed; this is the result that turns SBC into a calibration test.

**Theorem 3.1** (Rank uniformity under calibration)**.** *Let $\tilde{\theta} \sim \pi(\theta)$ and $\tilde{y} \sim \pi(y \mid \tilde{\theta})$, and let $\theta_1, \ldots, \theta_L$ be independent draws from the exact posterior $\pi(\theta \mid \tilde{y})$. For any one-dimensional summary $f$, the rank $r$ of equation* (3.3) *satisfies*

$$r \sim \mathrm{Uniform}\{0, 1, \ldots, L\}. \tag{3.4}$$

*Proof.* By the self-consistency identity (3.1) every posterior draw $\theta_\ell$ shares the marginal distribution $\pi(\theta)$ of $\tilde{\theta}$. The $L + 1$ values $(\tilde{\theta}, \theta_1, \ldots, \theta_L)$ are therefore exchangeable, so $\tilde{\theta}$ is equally likely to occupy any of the $L + 1$ order positions among the draws, and its rank is uniform on $\{0, \ldots, L\}$. □

In the limit $L \to \infty$ the normalised rank converges to the posterior quantile of the true value,

$$\frac{r}{L} \;\xrightarrow{L\to\infty}\; u \;=\; \Pr_{\theta\sim\pi(\cdot\mid\tilde{y})}\Big[\, f(\theta) < f(\tilde{\theta}) \,\Big], \qquad u \sim \mathrm{Uniform}(0, 1), \tag{3.5}$$

which is the posterior-quantile criterion of Cook, Gelman, and Rubin (2006); simulation-based calibration is its discrete, rank-based and Markov-chain-robust counterpart.

Over $N$ independent iterations the ranks are collected into a histogram of $B$ bins. Each iteration contributes exactly one rank, and under (3.4) that rank is equally likely to land

in any of the $B$ bins, independently of the others. The count $c_b$ in a given bin is therefore a sum of $N$ independent Bernoulli trials, each a success (the rank landing in bin $b$) with probability $1/B$, which is by definition a binomial,

$$c_b \sim \text{Binomial}\Big(N, \tfrac{1}{B}\Big), \qquad \mathbb{E}[c_b] = \frac{N}{B}, \tag{3.6}$$

so every bar should sit at the same expected height $N/B$ up to binomial sampling noise, and a flat histogram lying inside the band $\Big[\,\text{qbinom}(\tfrac{\alpha}{2}, N, \tfrac{1}{B}),\ \text{qbinom}(1-\tfrac{\alpha}{2}, N, \tfrac{1}{B})\Big]$ (with $\alpha = 0.01$, the grey band of Figure 3.4) is consistent with calibration. Characteristic departures instead diagnose specific failures: a $\cup$ shape, with mass piling at the two extremes 0 and $L$, signals posteriors that are too *narrow* (over-confident), because the true value then falls in the tails; a $\cap$ shape, with mass in the centre, signals posteriors that are too *wide* (under-confident); and a monotone slope signals a location bias. Two practical points matter. First, because Markov-chain draws are autocorrelated, the $L$ retained draws are thinned until approximately independent, as otherwise the ranks depart from uniformity even for an exact posterior. Second, the prior used to generate the data must coincide *exactly* with the one used to fit; the refinement of Modrák et al. (2023) shows, moreover, that augmenting the parameters with data-dependent test quantities makes the check sensitive to failure modes that the parameter ranks alone can miss.

We ran SBC on the same model and tree. As the method requires, and unlike the recovery experiment above, the ground truth here is drawn from the *exact* model prior, with no clamping, so that the prior used to generate the data and the prior used to fit it coincide; replicates that failed to converge are discarded, since the rank of a non-converged chain is meaningless. Figure 3.4 shows the resulting rank histograms, one per parameter, with the grey band giving the 99% interval expected under uniformity. Of the 1000 simulations, 966 converged and were retained.[1] Across the board the histograms are consistent with uniformity. For the global parameters ($\bar{w}$, $\sigma_{\log w}$, $\omega$) and for the six event times the bars scatter

[1] A small fraction of transitions (around 2% on average) ended in divergences; since every retained fit satisfied $\hat{R} < 1.1$ this does not affect the rank statistics, although it suggests that a higher `adapt_delta` would be prudent in production runs.

around the expected level with no systematic ∪, ∩ or sloped pattern, so the corresponding posteriors are neither over- nor under-confident and show no detectable bias. The per-clone growth histograms ($w_{\text{node}}$) are visibly noisier, and one or two bins reach the edge of the band. This is about what one would expect by chance from the roughly 280 bins inspected at the 99% level, and is unsurprising given the weaker identifiability of growth; the histograms still show no coherent departure from uniformity. Together with the recovery experiment, this is evidence that the sampler and the model implementation are faithful and that the uncertainties the model reports are calibrated.

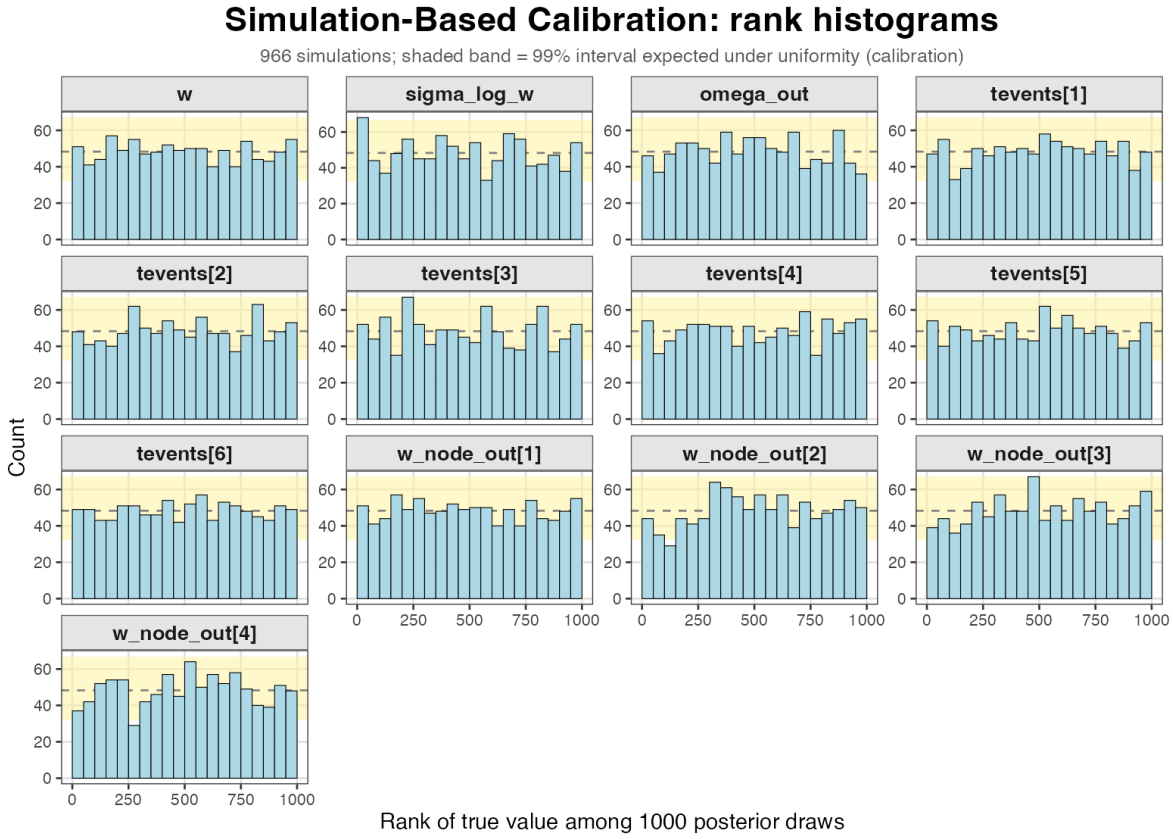



Figure 3.4: Simulation-based calibration: rank of the true value among the (thinned) posterior draws, aggregated over simulations, for each model parameter. Under a calibrated inference the ranks are uniform; the grey band is the 99% interval expected under uniformity. A ∪ shape would indicate over-confidence (intervals too narrow), a ∩ shape under-confidence (too wide), and a slope a bias.

### 3.1.3 Robustness checks

A trustworthy inference should not hinge on choices the analyst makes somewhat arbitrarily, nor collapse when the data grow scarcer. The robustness checks probe exactly this: we vary, one quantity at a time, the inputs the model is most exposed to, and watch whether the recovery of the event times holds up. Four sweeps are run under the ideal DGP, testing the quantities most open to question:

- the width of the prior on the clock-to-growth factor $\omega$;
- the rate $\lambda$ of the prior on the random-walk step $\sigma_{\log w}$;
- the number of sequenced cells (the sampling depth);
- the dispersion $\varphi$ of the negative-binomial abundance likelihood.

For each sweep we record, for every event, three quantities averaged over replicates: the absolute bias (the distance of the posterior median from the truth), the 95% coverage (how often the credible interval contains the truth), and the width of that credible interval, that is the length of the 95% interval, which measures how sharp or uncertain the estimate is. A good inference keeps coverage near the nominal 0.95 while making the width as small as the data allow; an over-tight prior, for instance, narrows the width but at the cost of coverage. The four figures below report one sweep each, followed by its reading.

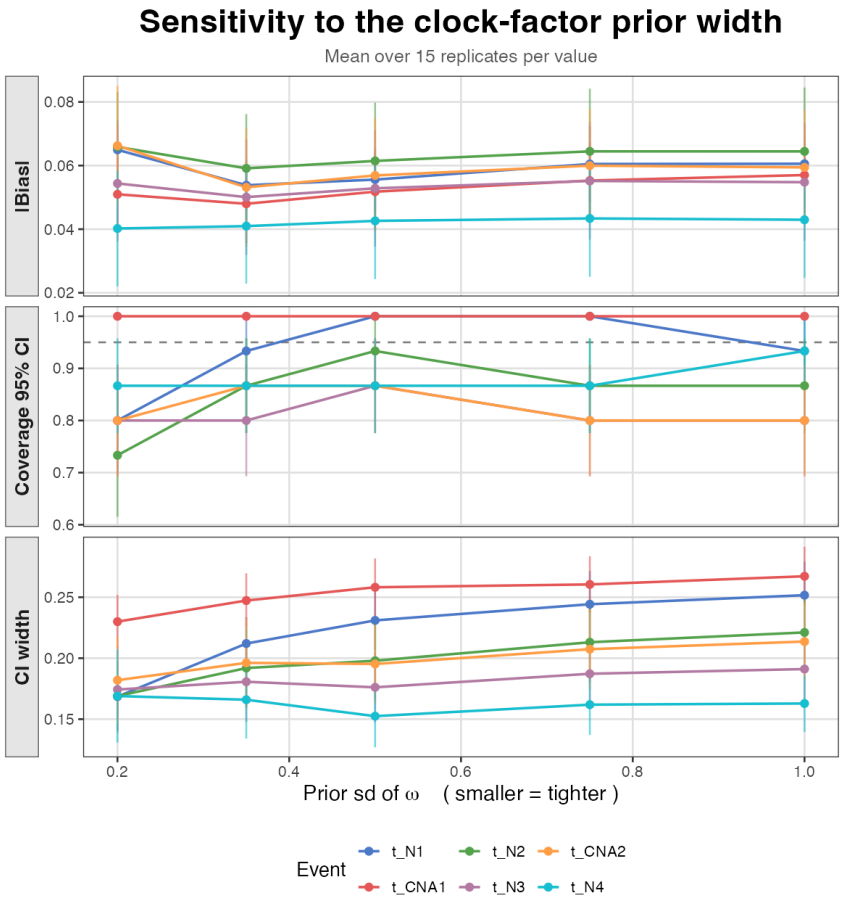



Figure 3.5: Sensitivity to the width of the clock-factor prior. Per-event absolute bias, 95% coverage and credible-interval width as the prior sd of $\omega$ is varied (mean over replicates; error bars are standard errors). An over-tight prior under-covers; the dashed line marks the nominal 0.95 coverage.

The width of the $\omega$ prior is the one choice that genuinely matters (Figure 3.5). An over-tight prior (sd $= 0.2$) yields over-confident intervals that under-cover for several events; from about sd $= 0.5$ onwards the coverage returns close to nominal, at the modest price of slightly wider intervals.

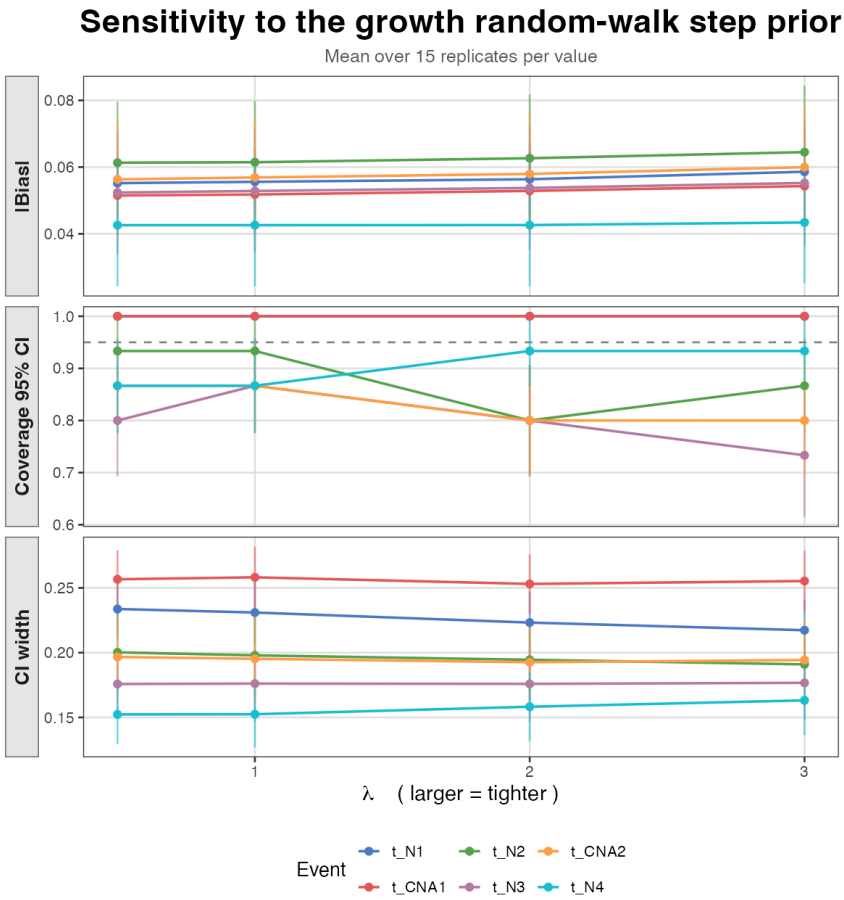



Figure 3.6: Sensitivity to the prior on the growth random-walk step, $\sigma_{\log w} \sim \text{Exponential}(\lambda)$. Bias and coverage are largely insensitive to $\lambda$ on this topology.

The prior on the growth random-walk step has a much milder effect (Figure 3.6): on this topology bias and coverage are essentially flat across $\lambda$, and the chosen $\lambda = 1$ is the value that best balances the competing demands of chain-like and branching trees.

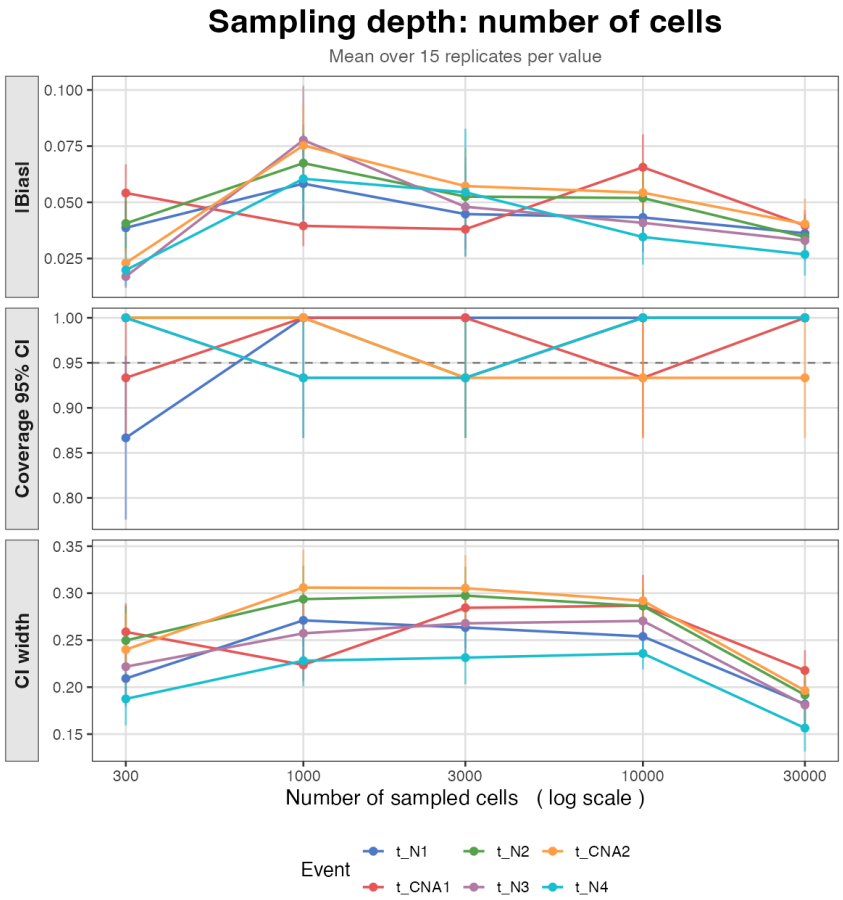



Figure 3.7: Effect of sampling depth. As the number of sequenced cells grows, the bias falls and the credible intervals tighten while coverage remains close to nominal.

Sampling depth behaves exactly as one would hope (Figure 3.7): more cells lower the bias and tighten the intervals while coverage stays close to nominal, a sign that the inference puts the extra data to good use instead of growing overconfident.

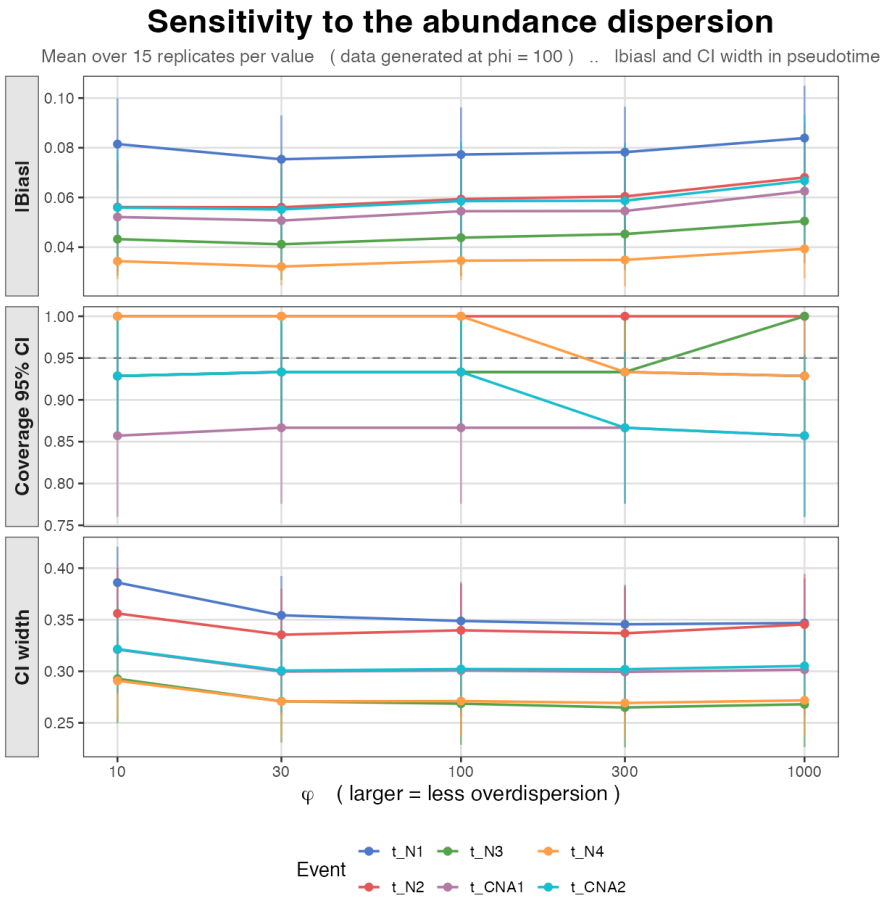



Figure 3.8: Sensitivity to the dispersion $\varphi$ of the negative-binomial abundance likelihood (data generated at $\varphi = 100$). Event-time bias and coverage are flat across two orders of magnitude of $\varphi$; only very strong overdispersion ($\varphi = 10$) widens the credible intervals.

The dispersion $\varphi$ of the abundance likelihood matters less still (Figure 3.8): event-time recovery barely moves as $\varphi$ ranges over two orders of magnitude, and only at the most overdispersed setting ($\varphi = 10$, the noisiest abundances) do the intervals widen a little, with bias and coverage flat throughout. This fits the rest of the chapter: the timing is carried by the mutation counts, so how noisy one assumes the abundances to be barely reaches the event times.

These checks point the same way. Under its own assumptions the model recovers event times accurately and with calibrated uncertainty, it passes simulation-based calibration, and it is robust both to reasonable changes in the priors and to the amount of data available.

The one limitation that surfaces, the residual uncertainty on per-clone growth, is a property of single-snapshot data rather than a flaw in the implementation, and it sets the bar against which the ProCESS experiments in the rest of this chapter should be judged.

## 3.2 ProCESS: a linear four-clone tumour

The ideal-DGP experiments show what the model can do under its own assumptions; the real test is whether those conclusions survive when the data are produced by a different mechanism. ProCESS is a spatial simulator of clonal tumour growth: cells divide, mutate and acquire copy-number alterations on a two-dimensional lattice, and a region of the final tumour is sequenced (Caravagna and Casagrande 2024).

We study two scenarios, a four-clone linear chain and a six-clone branching tree, both sampled by a localized biopsy ($DLP^{+}$): a contiguous patch of tissue that over- and under-represents clones and so distorts the abundance signal the model relies on. This is the regime that matters in practice, since a real biopsy is never a proportional draw from the clone simplex; the well-specified, proportional-like case is the one already covered by the ideal-DGP analysis above. Any degradation seen here therefore measures robustness to a doubly adverse setting at once: a different growth mechanism *and* a distorted sample. The two trees are summarized in the cards opening the respective sections; for each we show the simulated tissue with the biopsied region marked, the reconstructed mutation phylogeny, and the posteriors of the event pseudotimes, the clone growth rates and the mutation counts against the ground truth. The bias of the event-time and growth-rate recovery is summarised, for both scenarios, in Table 3.1. Event times are on the unit pseudotime scale, $t = 0$ the tumour origin and $t = 1$ the moment of sequencing.

## Linear chain (4 clones)

Nnodes=4 | Nevents=6 | Nequations=8

**Topology** (linear, 4 sampled, 0 hidden)

`Clone_A` ⟶ `Clone_B` ⟶ `Clone_C` ⟶ `Clone_D`

| | |
|---|---|
| **Sampled** | Clone_A, Clone_B, Clone_C, Clone_D |
| **Sampling** | localized biopsy ($DLP^+$) |
| **Stan parents** | (0, 1, 2, 3) |
| **number_of_events** | (2, 1, 2, 1) |
| **CNAs** | CNA_A on Clone_A – chr12, 40 Mb |
| | CNA_C_hidden in Clone_C – chr5, 20 Mb |
| **Growth rates** | $w_A$=0.0493 $w_B$=0.0983 $w_C$=0.1350 $w_D$=0.1883 |
| **Equal rates** | none |

The localized biopsy of this tumour returned only 192 cells, very unevenly split across the four clones: 10 from the founder $A$, 98 from $B$, 47 from $C$ and 37 from $D$. On the simulator's clock the tumour is sequenced at $T_f \approx 86$, and the ground-truth births are $t_B = 0.38$, $t_C = 0.53$, $t_D = 0.74$ in pseudotime (real times $\approx$ 33, 46 and 64), with the two root-clone events pinned at the origin. The clone growth rates and copy-number events are those listed in the card above; the recovered quantities are collected in Table 3.1.

The four-clone chain is the gentler of the two trees. The biopsy (Figure 3.9) over- and under-represents the clones, yet the inference holds up. The event pseudotimes are recovered with a mean absolute error of 0.020, the only two events left uncovered being those on the root clone A, whose true time is essentially zero: pinned against the boundary $t = 0$, a strictly positive posterior cannot quite reach them, so these are boundary artefacts rather than genuine misses (Figure 3.11). The growth rates are recovered to about 11% with all four intervals covering the truth (Figure 3.12).

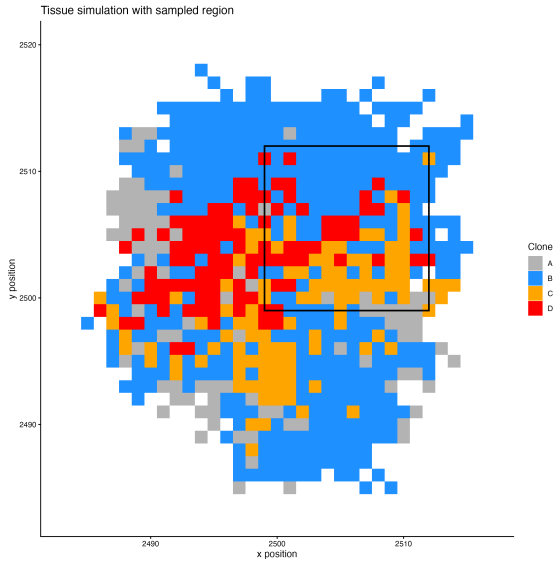



Figure 3.9: The simulated four-clone tumour, each colour a clone; the black square marks the localized biopsy (DLP$^+$), a contiguous patch of tissue that over- and under-represents clones and so distorts the abundance signal.




Figure 3.10: Mutation phylogeny reconstructed from the biopsy sample.

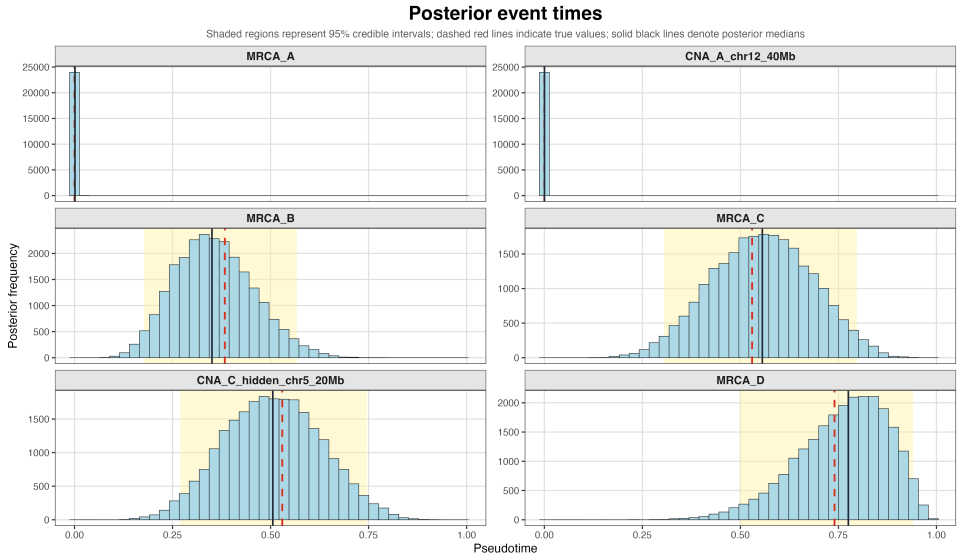



Figure 3.11: Posterior event pseudotimes against the ground truth (dashed) for the four-clone scenario; only the two root-clone events, fixed at $t \approx 0$, fall outside their interval.

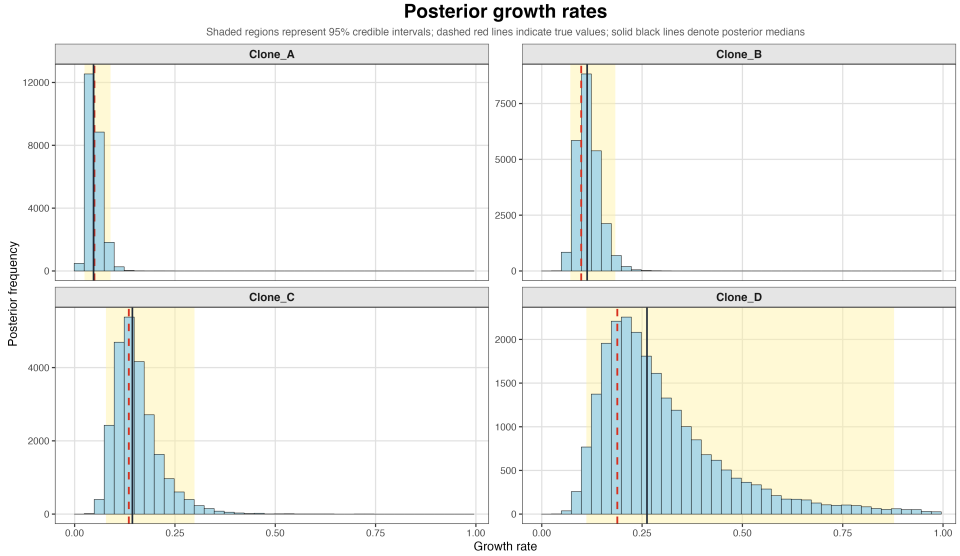



Figure 3.12: Posterior clone growth rates for the four-clone scenario under localized biopsy sampling.

The mutation counts are intrinsic to each branch rather than to how the tissue is sampled, so the biopsy distortion does not reach them. The posterior-predictive check is satisfied on every one of the eight equations (Figure 3.13): the abundance distortion is absorbed by the model without leaking into the mutation likelihood.

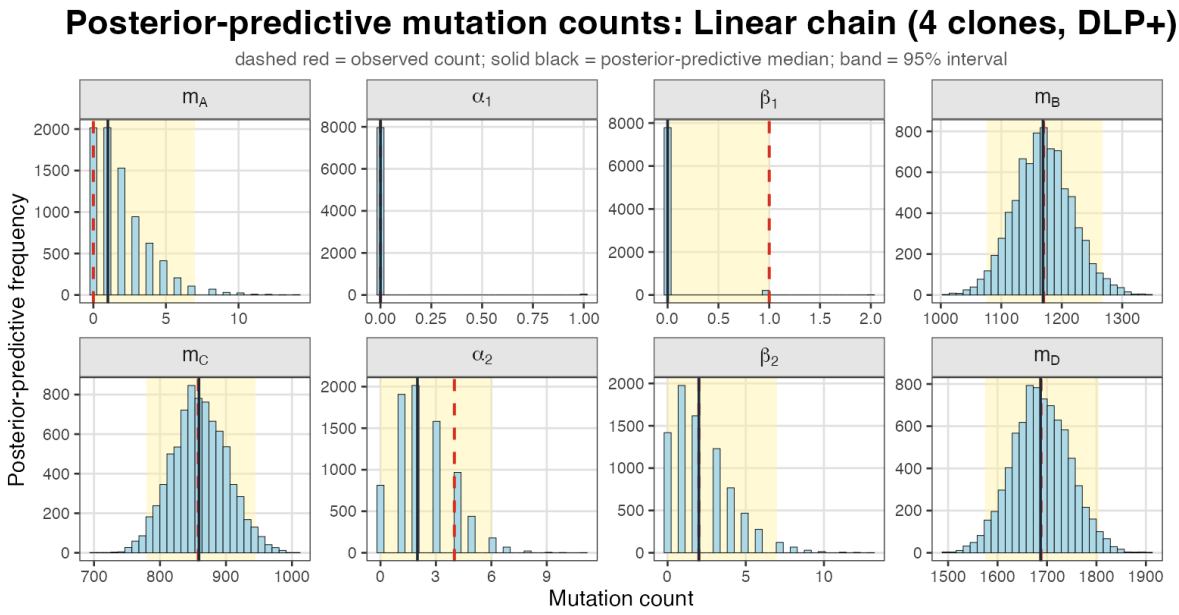



Figure 3.13: Posterior-predictive distribution of the mutation counts for the four-clone scenario. Each panel is one mutation equation; the dashed red line is the observed count, the solid line the predictive median, and the band the 95% predictive interval. All eight observed counts fall inside their intervals.

Taken together, the four-clone scenario is a clean pass: despite the distorting biopsy, the event times, the growth rates and the mutation counts are all recovered within their credible intervals (Table 3.1). The gentlest ProCESS test poses no real difficulty for the model.

## 3.3 ProCESS: a branching six-clone tumour

**Branching tree (6 clones)** Nnodes=6 | Nevents=9 | Nequations=12

**Topology** (branching, 6 sampled, 0 hidden)

Clone_A ⟶ Clone_B ⟶ Clone_C ⟶ Clone_E

↘ Clone_D ⟶ Clone_F

| | |
|---|---|
| **Sampled** | Clone_A, Clone_B, Clone_C, Clone_D, Clone_E, Clone_F |
| **Sampling** | localized biopsy ($DLP^+$) |
| **Stan parents** | (0, 1, 2, 2, 3, 4) |
| **number_of_events** | (2, 1, 2, 2, 1, 1) |
| **CNAs** | CNA_A on Clone_A – chr12, 40 Mb |
| | CNA_C_hidden in Clone_C – chr5, 20 Mb |
| | CNA_D_hidden in Clone_D – chr18, 20 Mb |
| **Growth rates** | $w_A$=0.0493 $w_B$=0.0920 $w_C$=0.0946 |
| | $w_D$=0.0969 $w_E$=0.1879 $w_F$=0.1165 |
| **Equal rates** | none |

The biopsy of the six-clone tumour returned 1369 cells, dominated by the leaf clone $E$ (883) and very sparse for $C$ (only 8): the strongly skewed composition a focal sample produces in practice (107, 267, 8, 37, 883, 67 for clones $A$–$F$). Sequencing is at $T_f \approx 141$ on the simulator's clock; the clones are born at pseudotimes $t_B = 0.29$, $t_C = 0.60$, $t_D = 0.72$, $t_E = 0.74$, $t_F = 0.77$ (real times $\approx$ 41, 85, 101, 104 and 108), and copy-number events sit on $A$, $C$ and $D$. Growth rates and CNAs are those of the card above; recovery is summarised in Table 3.1.

The six-clone branching tree, carrying three copy-number events, is the harder problem, and it is here that the identifiability limit anticipated by the ideal-DGP analysis becomes visible. The pseudotimes are recovered with a mean absolute error of 0.056, larger than on the four-clone chain, with the root events again among those left uncovered (Figure 3.16).

The growth rates carry a larger relative error, around 22% (Figure 3.17): exactly the single-snapshot floor predicted on a deeper tree. The intervals still cover the truth for all six clones, so rather than settling on a wrong value the model widens its uncertainty, as under the ideal DGP.




Figure 3.14: The simulated six-clone branching tumour, each colour a clone; the black square marks the localized biopsy (DLP$^{+}$) used to sample it.

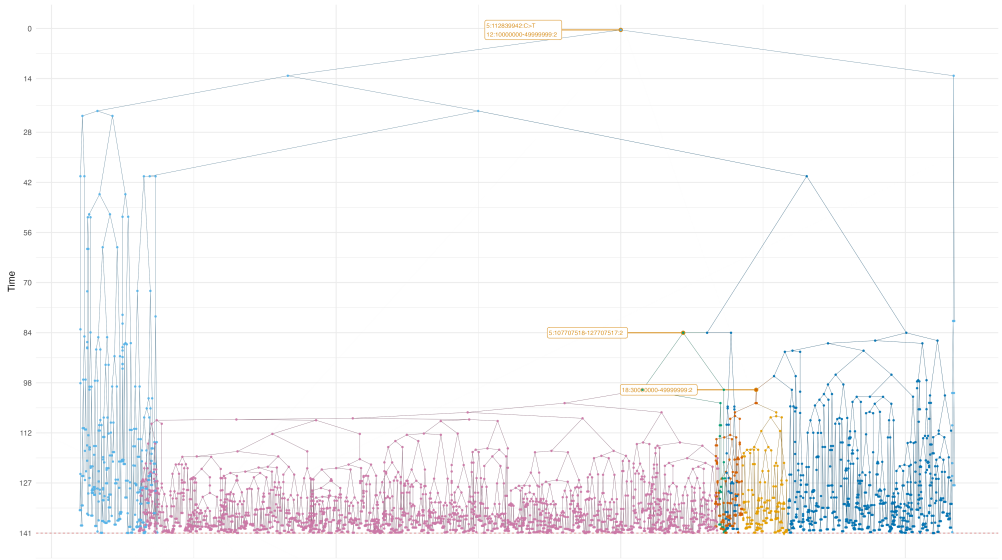



Figure 3.15: Mutation phylogeny reconstructed from the biopsy sample of the six-clone tumour.

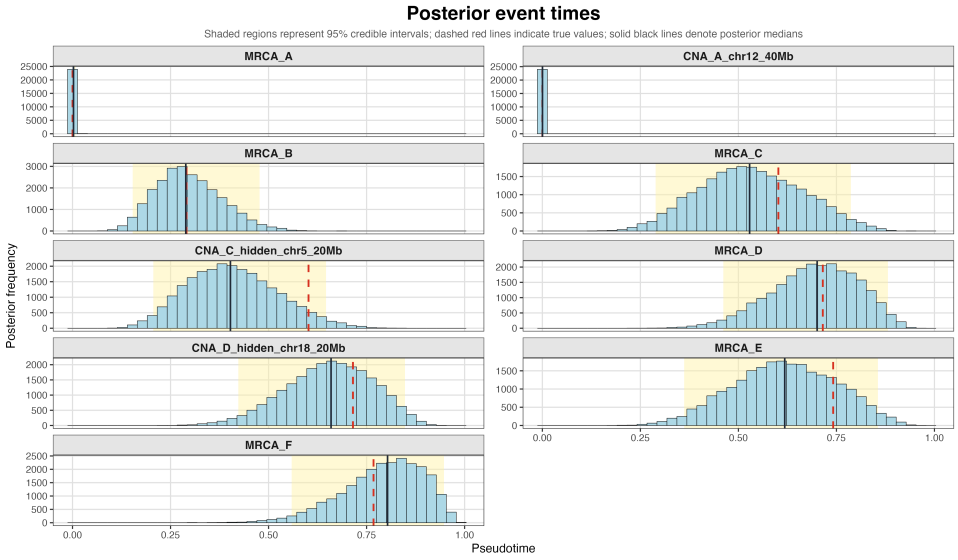



Figure 3.16: Posterior event pseudotimes for the six-clone scenario under localized biopsy sampling.

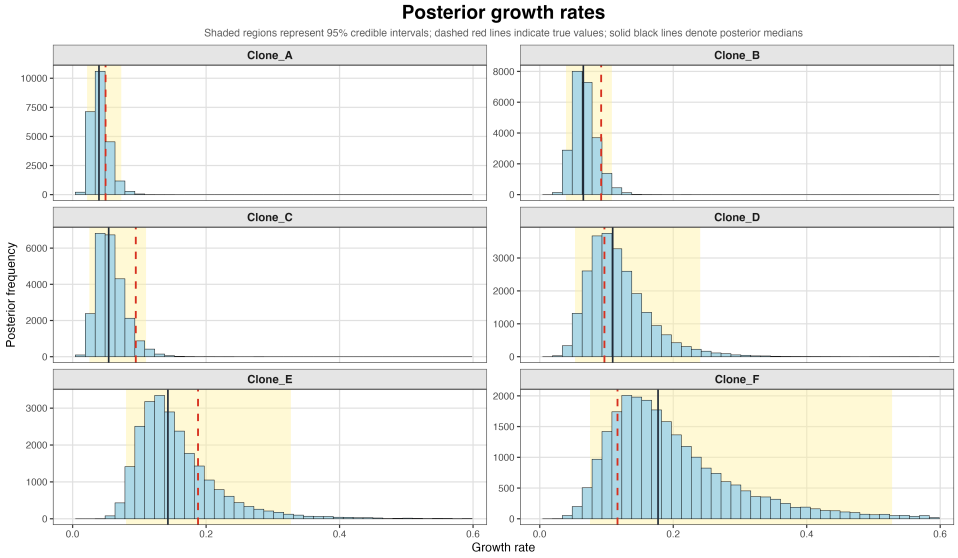



Figure 3.17: Posterior growth rates for the six-clone scenario under localized biopsy sampling; coverage is retained for all six clones.

The mutation counts are reproduced on the harder tree too: the observed value lies inside the 95% posterior-predictive band for eleven of the twelve equations (Figure 3.18). The lone exception is a low-count branch, where the discreteness of small Poisson counts makes the interval narrow and a handful of observed mutations sit just above a predictive distribution

concentrated near zero; the high-count branches, in the hundreds to nearly two thousand, are matched almost exactly.

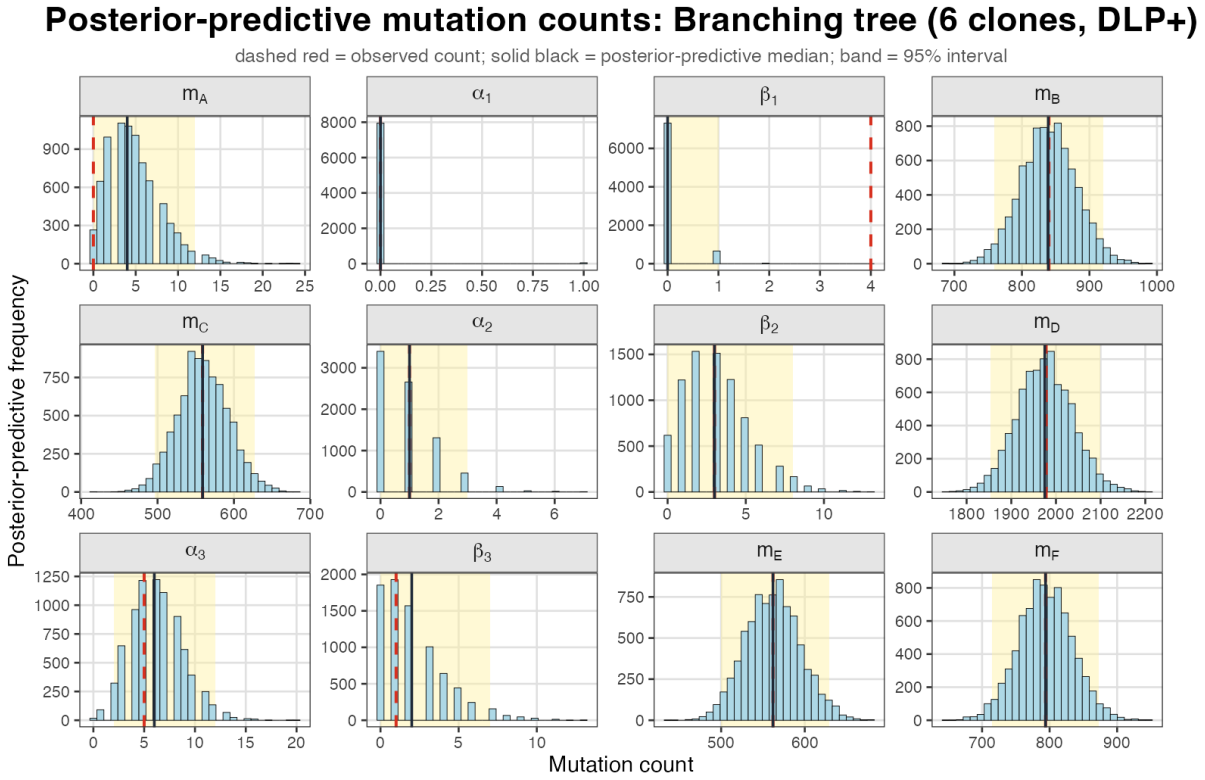



Figure 3.18: Per-equation posterior-predictive distribution of the mutation counts for the six-clone scenario under localized biopsy. Each panel is one mutation equation; the dashed red line is the observed count, the solid line the predictive median, and the band the 95% predictive interval. Eleven of the twelve observed counts fall inside their intervals; the exception is the low-count branch (Eq 03).

The six-clone scenario splits the verdict: the event times and the mutation counts are reproduced, while the growth rates lose precision though not calibration.

Table 3.1 collects the recovery metrics for the two ProCESS scenarios, reported separately for the event pseudotimes and the clone growth rates. Each block holds three summaries, averaged over the events or clones of that scenario. The mean absolute bias (MAB) is the average distance between the posterior median and the truth, in the same units as the quantity (pseudotime on $[0, 1]$ for the times, growth units for the rates). The root-mean-square error (RMSE) is the square root of the average squared error: it weights large misses

more heavily than MAB, so a gap between the two columns flags a few events estimated much worse than the rest. The median relative bias (Rel. bias) is the typical *signed* error as a fraction of the true value, so its magnitude says how large the error is in relative terms and its sign says whether the inference tends to over-estimate (positive) or under-estimate (negative). Lower MAB and RMSE are better, and a relative bias near zero means no systematic over- or under-estimation. The pseudotimes are on a common $[0, 1]$ scale, so their MAB and RMSE are directly comparable between the two scenarios.

Table 3.1: Posterior bias of the inference for the two ProCESS scenarios, reported separately for the event pseudotimes and the clone growth rates. MAB is the mean absolute bias, RMSE the root-mean-square error, and Rel. bias the median relative bias; all three are averaged over the events or clones of each scenario. Lower MAB and RMSE are better, and the sign of the relative bias gives the direction of the error.

| | Event pseudotime | | | Growth rate | | |
|---|---|---|---|---|---|---|
| Scenario | MAB | RMSE | Rel. bias | MAB | RMSE | Rel. bias |
| Linear chain (4 clones) | 0.020 | 0.024 | 0.001 | 0.025 | 0.038 | 0.108 |
| Branching tree (6 clones) | 0.056 | 0.085 | -0.078 | 0.033 | 0.037 | -0.220 |

The growth-rate columns make the pattern explicit: the median relative bias rises from about 11% on the four-clone chain to about 22% on the deeper six-clone tree, while the event-time error stays small (MAB $\leq 0.06$ in pseudotime) in both. It is the depth of the phylogeny, not the way the tissue is sampled, that limits how well the per-clone growth can be pinned down.

# 4. Discussion and Conclusions

## 4.1 Summary of the Work

This thesis develops and tests a Bayesian model for reconstructing the temporal order of the genomic events in a growing tumour. From a single snapshot of single-cell sequencing data, the model infers the timing of clone-birth and copy-number events together with the growth rate of each clone. Four pieces of information feed into the same latent timeline: the number of somatic mutations on each branch, the way mutations in a copy-number region split into those that arose before and after the event, the relative abundance of the sampled clones, and the topology of the phylogenetic tree. The growth rates are given a hierarchical random-walk prior on the log scale, which lets related clones share information while still allowing distinct clones to differ, and the event times are parametrised by stick-breaking so that the tree ordering constraints are always satisfied.

A large part of the work went into validating the model, in two stages. The model was first fitted to data generated by its own process, the well-specified case, where parameter recovery, simulation-based calibration and a set of sensitivity checks separate the behaviour of the inference from any question of model fit. It was then applied to synthetic data from ProCESS (Caravagna and Casagrande 2024), a spatial tumour simulator whose biology only partly matches the modelling assumptions, so that the second stage measures robustness to misspecification.

## 4.2 Main Findings

The two stages give a consistent picture. Under the well-specified process the event times are recovered accurately and, just as important, with calibrated uncertainty: their credible intervals reach close to nominal coverage, and simulation-based calibration (Talts et al. 2018) gives rank histograms compatible with uniformity for every parameter, so the

posteriors are neither over- nor under-confident. The growth rates behave differently. Their point estimates have a larger relative error, but their intervals still cover the truth, because the model widens the uncertainty rather than settling on a value the data cannot determine. This is the identifiability limit of a single snapshot, which constrains the product of a growth rate and an elapsed time much better than either factor on its own.

The same conclusions hold on the ProCESS data. Both scenarios are read from a localized biopsy, the adverse case where a contiguous patch of tissue distorts the clone proportions, so this is a deliberately hard test rather than an idealised one. Even then the event pseudotimes are recovered with small absolute error (Table 3.1), and the only events that fall outside their intervals are the root events fixed at the origin of time, a boundary effect rather than a real failure. The growth rates are recovered well on the four-clone chain and lose precision, though not calibration, on the deeper six-clone tree, in line with what the well-specified analysis predicted. Because both trees are read under the same biopsy, the comparison points to the cause: what limits the per-clone growth is the depth of the lineage, the single-snapshot floor once more, not the way the tissue is sampled.

## 4.3 Limitations

A few limitations should be stated plainly. The first is the single-snapshot identifiability of growth: from one time point the model can give calibrated intervals on the per-clone growth rates, but it cannot make them arbitrarily narrow, and on deep lineages the residual uncertainty is large. This floor is a property of the data, not of the implementation, and the prior cannot remove it: along the unidentified direction the likelihood is flat, so the posterior simply follows the prior. A vague, honest prior therefore leaves the floor in place, whereas a tighter one would not solve the problem but merely assert an answer, narrow yet possibly wrong and over-confident, a miscalibration that simulation-based calibration would expose; genuinely shrinking the growth intervals takes more data, not a sharper prior. Second, the inference treats the phylogenetic tree and the assignment of copy-number events to branches as known inputs; these come from upstream reconstruction, and errors in them

would propagate into the timing estimates. Third, ProCESS is itself a model: real single-cell data carry noise that is absent here, such as sequencing and copy-number-calling error, doublets and allele dropout, so the robustness shown in this chapter is necessary but not sufficient evidence of real-world performance. Finally, a small fraction of the extreme prior draws produced divergent transitions; these did not affect the validation, but they suggest that a more conservative sampler adaptation would be sensible for routine use.

## 4.4 Future Work

The clearest next step follows from the main limitation. The growth–time degeneracy arises because the tumour is observed only once, so the natural remedy is a *longitudinal* design that samples it at more than one time. With serial samples the ratio of a clone's size between two timepoints measures its growth rate on its own, independently of the elapsed time, and so separates the product $w \cdot t$ into its factors. One caveat sharpens the requirement: single-cell composition alone is invariant to the overall scale, so the timepoints must also carry information on the absolute tumour burden, such as a total cell count or a circulating-tumour-DNA load, for the growth rates to be pinned down. Given such data, the per-clone growth that a single snapshot leaves uncertain would become tightly identified. A second, methodological direction is to strengthen the calibration check with the data-dependent test quantities of Modrák et al. (2023), which catch problems that the parameter ranks alone miss, and to let the molecular clock vary by branch to absorb rate heterogeneity. A third is to infer the tree and the copy-number assignments jointly with the timings, instead of fixing them, so that their uncertainty is carried through. Finally, replacing exponential growth with a spatial or density-dependent model would bring the model closer to simulators such as ProCESS, and eventually to real tissue.

## 4.5 Conclusion

The results show that the evolutionary timing of a tumour can be reconstructed from single-cell data with a Bayesian model that is accurate and clear about its own uncertainty. Event times are recovered reliably and with calibrated intervals; growth rates are recovered as far as a single snapshot allows, with the residual uncertainty reported rather than hidden; and the inference is robust to the way the tissue is sampled. The one quantity a single snapshot cannot fully pin down, the absolute scale of the per-clone growth rates, also points to the clearest way forward: a *longitudinal* design, in which the same tumour is sequenced at more than one time and anchored to an absolute measure of its burden, such as a total cell count or a circulating-tumour-DNA load, would supply precisely the information the single snapshot lacks. The ratio of a clone's size between two timepoints measures its growth directly, independently of the elapsed time, so serial data would turn the growth rates from quantities the snapshot can only bound into sharply identified ones, while the timing machinery developed here carries over unchanged. The staged validation, from the well-specified process through simulation-based calibration to ProCESS, is part of the contribution, since it gives a way to trust the inferences before any real data are used. The model is a solid starting point for application to patient single-cell sequencing, where calibrated timing of clonal events could help us understand how, and how fast, tumours evolve.

# A. Appendix - Computational Implementation

All models in this thesis are implemented in the Stan probabilistic programming language and fitted with its dynamic Hamiltonian Monte Carlo sampler (the No-U-Turn sampler) through the `cmdstanr` interface. The source code, together with the simulation and validation scripts, lives in the project repository (`stan_models/` and `robustness_checks/`). Two design choices are shared by every model. Event times are never sampled directly: they are built from auxiliary variables $u \sim \text{Uniform}(0, 1)$ by stick-breaking, so that the tree ordering constraints $t_{\text{pa}(i)} < t_i$ hold by construction and the sampler explores an unconstrained space. Clone abundances enter through a negative-binomial likelihood on the simplex of terminal clone masses, which makes the abundance signal scale-free and robust to the absolute number of sequenced cells. Numerical guards (`fmin` caps inside the exponentials) keep the rates finite even for extreme parameter draws.

## A.1 Stan Implementation of the Basic Model

The basic model introduced in Chapter 2 treats the clone growth rates as known and infers only the event pseudotimes. Computationally it is the complete model of Appendix A.2 with two simplifications: the growth rates $w_1, \ldots, w_K$ are supplied as data rather than given a prior, so the entire random-walk block (`w_bar`, `sigma_log_w` and `delta_log_w`) and the global clock factor $\omega$ are removed, leaving the stick-breaking auxiliaries $u$ as the only parameters. The mutation, copy-number and abundance likelihoods are otherwise unchanged, with each clone's known $w_i$ substituted wherever the complete model uses its inferred value. Because growth is fixed, the basic model is fully identified; it serves, as discussed in Chapter 2, as the stepping stone whose deterministic counterpart is derived in Appendix B.1. The listing is given below.

```
// ============================================================================
// BASIC MODEL -- known clone growth rates, only event times inferred.
```

```
//   The per-clone growth rates w_node are supplied as data; the model infers the
//   event pseudotimes via nested-uniform stick-breaking.
//   mutations:  m ~ Poisson(w_i * mu * dt * l * coeff * dt_interval)
//   abundance:  N_i ~ NegBinomial(N * p_i, phi),  p_i from exp(w_i*dt*(1-t_i))
//   This is the complete model with the growth random walk and the clock factor
//   removed; with growth fixed the event times are fully identified.
// ============================================================================
functions {
  real positive_rate(real rate) { return fmax(rate, 1e-9); }
}

data {
  real Tf_real;
  real t0_real;
  real<lower=0> mu;
  real<lower=0> phi;

  int<lower=1> Nnodes;
  array[Nnodes] int<lower=0, upper=Nnodes> parents;
  array[Nnodes] int<lower=1> number_of_events;
  array[Nnodes] int<lower=1> idx_first_event;
  array[Nnodes] real<lower=0> w_node;                  // KNOWN clone growth rates

  int<lower=1> Nevents;
  array[Nevents] int<lower=0, upper=1> karyotypes;     // 0 = MRCA, 1 = CNA
  array[Nevents] real<lower=0> lengths;
  array[Nevents] int<lower=1> idx_first_mutation;

  int<lower=1> Nequations;
  array[Nequations] real<lower=0> coeff;               // allelic-copy coefficients
  array[Nequations] int<lower=0> mutations;            // observed mutation counts

  int<lower=0> Ncells;
  array[Nnodes] int<lower=0> Ncells_per_clone;
}

parameters {
  vector<lower=0, upper=1>[Nevents] u;                 // stick-breaking auxiliaries
}

transformed parameters {
  real<lower=0> delta_t = Tf_real - t0_real;
  array[Nevents] real<lower=0, upper=1> tevents;
```

```
  vector<lower=0>[Nnodes] terminal_clone_mass;
  simplex[Nnodes] clone_prob_cells;

  // --- event times: nested-uniform stick-breaking ---
  for (idx in 1:Nevents) tevents[idx] = 0;
  for (node in 1:Nnodes) {
    int idx0 = idx_first_event[node];
    int p = parents[node];
    for (event in 1:number_of_events[node]) {
      int idx = idx0 + event - 1;
      if (p == 0) {
        tevents[idx] = (event == 1) ? u[idx] : u[idx] * tevents[idx0];
      } else {
        real t_parent = tevents[idx_first_event[p]];
        tevents[idx] = (event == 1)
          ? t_parent + u[idx] * (1.0 - t_parent)
          : t_parent + u[idx] * (tevents[idx0] - t_parent);
      }
    }
  }

  // --- abundance: exponential mass minus the children's contribution ---
  for (node in 1:Nnodes) {
    real t_node = tevents[idx_first_event[node]];
    terminal_clone_mass[node] = exp(fmin(w_node[node] * delta_t * (1.0 - t_node), 200));
  }
  for (child in 2:Nnodes) {
    int p = parents[child];
    real t_child = tevents[idx_first_event[child]];
    terminal_clone_mass[p] -= exp(fmin(w_node[p] * delta_t * (1.0 - t_child), 200));
  }
  for (node in 1:Nnodes) {
    if (terminal_clone_mass[node] < 1e-12) terminal_clone_mass[node] = 1e-12;
  }
  clone_prob_cells = terminal_clone_mass / sum(terminal_clone_mass);
}

model {
  // u ~ Uniform(0,1) implicit; the growth rates w_node are known (data)

  // mutation likelihood (Poisson), rate proportional to the known growth w
  for (node in 1:Nnodes) {
    int idx0 = idx_first_event[node];
```

```
    int p = parents[node];
    real t_p = (p == 0) ? 0.0 : tevents[idx_first_event[p]];

    for (event in 1:number_of_events[node]) {
      int idx = idx0 + event - 1;
      if (event == 1) {                                  // MRCA -> diploid clonal m
        int idx_m = idx_first_mutation[idx0];
        real rate_m = w_node[node] * mu * delta_t * lengths[idx0]
                      * coeff[idx_m] * (tevents[idx0] - t_p);
        target += poisson_lpmf(mutations[idx_m] | positive_rate(rate_m));
      } else {                                           // CNA -> alpha (pre) + beta (post)
        int idx_a = idx_first_mutation[idx];
        int idx_b = idx_first_mutation[idx] + 1;
        real t_cna = tevents[idx];
        real rate_a = w_node[node] * mu * delta_t * lengths[idx]
                      * coeff[idx_a] * (t_cna - t_p);
        real rate_b = w_node[node] * mu * delta_t * lengths[idx]
                      * coeff[idx_b] * (tevents[idx0] - t_cna);
        target += poisson_lpmf(mutations[idx_a] | positive_rate(rate_a));
        target += poisson_lpmf(mutations[idx_b] | positive_rate(rate_b));
      }
    }
  }

  // abundance likelihood (Negative Binomial)
  for (node in 1:Nnodes) {
    target += neg_binomial_2_lpmf(
      Ncells_per_clone[node] | positive_rate(Ncells * clone_prob_cells[node]),
      positive_rate(phi)
    );
  }
}

generated quantities {
  // posterior predictive replicated mutation counts (PPC on mutation numbers)
  array[Nequations] int mutations_rep;
  for (node in 1:Nnodes) {
    int idx0 = idx_first_event[node];
    int p = parents[node];
    real t_p = (p == 0) ? 0.0 : tevents[idx_first_event[p]];
    for (event in 1:number_of_events[node]) {
      int idx = idx0 + event - 1;
      if (event == 1) {
```

```
        int idx_m = idx_first_mutation[idx0];
        mutations_rep[idx_m] = poisson_rng(positive_rate(
          w_node[node] * mu * delta_t * lengths[idx0] * coeff[idx_m] * (tevents[idx0] - t_p)));
      } else {
        int idx_a = idx_first_mutation[idx];
        int idx_b = idx_first_mutation[idx] + 1;
        real t_cna = tevents[idx];
        mutations_rep[idx_a] = poisson_rng(positive_rate(
          w_node[node] * mu * delta_t * lengths[idx] * coeff[idx_a] * (t_cna - t_p)));
        mutations_rep[idx_b] = poisson_rng(positive_rate(
          w_node[node] * mu * delta_t * lengths[idx] * coeff[idx_b] * (tevents[idx0] - t_cna)));
      }
    }
  }
}
```

## A.2 Stan Implementation of the Complete Model

The complete model is the random-walk model of Section 2.2: it infers the event pseudotimes and the clone-specific growth rates jointly. Growth is given a hierarchical random walk on the log scale, with a root rate `w_bar` and per-branch increments `delta_log_w` of standard deviation `sigma_log_w`, and the molecular clock is coupled to growth through a single global factor $\omega$. Somatic mutations follow a Poisson law whose rate is proportional to this clock, the genomic length and the branch duration, with copy-number branches split into the pre- and post-event counts $\alpha$ and $\beta$. The priors are $\bar{w} \sim$ LogNormal, $\sigma_{\log w} \sim$ Exponential and $\omega \sim$ LogNormal($\log 1.25$, 0.5), the last deliberately tight to break the global-scale multimodality discussed in Chapter 2. Inference uses four chains with several thousand post-warm-up draws each, `adapt_delta` raised to 0.95 and a maximum tree depth of 13. The complete listing is given below.

```
// ============================================================================
// BASE MODEL -- random walk on growth, NO per-clone residual.
//   growth:     log w_i = log w_parent + sigma_log_w * z_i      (RW on the tree)
//   clock:      nu_i = omega * w_i                              (rigid, no residual)
//   mutations:  m ~ Poisson(nu_i * mu * dt * l * coeff * dt_interval)   (alpha uses
//               the clone's own w_node, NOT w_parent -- alpha/beta both on the stem)
```

```
//   abundance:  N_i ~ NegBinomial(N * p_i, phi),  p_i from exp(w_i*dt*(1-t_i))
//   event times: nested-uniform stick-breaking (uniform prior).

// ============================================================================
functions {
  real positive_rate(real rate) { return fmax(rate, 1e-9); }
}

data {
  real Tf_real;
  real t0_real;
  real<lower=0> mu;
  real<lower=0> phi;

  int<lower=1> Nnodes;
  array[Nnodes] int<lower=0, upper=Nnodes> parents;
  array[Nnodes] int<lower=1> number_of_events;
  array[Nnodes] int<lower=1> idx_first_event;

  int<lower=1> Nevents;
  array[Nevents] int<lower=0, upper=1> karyotypes;     // 0 = MRCA, 1 = CNA
  array[Nevents] real<lower=0> lengths;
  array[Nevents] int<lower=1> idx_first_mutation;

  int<lower=1> Nequations;
  array[Nequations] real<lower=0> coeff;               // allelic-copy coefficients
  array[Nequations] int<lower=0> mutations;            // observed mutation counts

  int<lower=0> Ncells;
  array[Nnodes] int<lower=0> Ncells_per_clone;

  real<lower=0> w_bar_prior_mean;
  real<lower=0> w_bar_prior_sd;
  real<lower=0> sigma_log_w_rate;
  real<lower=0> omega_prior_mean;   // prior mean of the clock-to-growth factor
  real<lower=0> omega_prior_sd;     // tightness; small -> nearly fixed
}

parameters {
  vector<lower=0, upper=1>[Nevents] u;                 // stick-breaking auxiliaries

  real<lower=1e-9, upper=1> w_bar;                     // root growth rate
  real<lower=0, upper=2> sigma_log_w;                  // RW step sd on log growth
```

```
  vector[Nnodes - 1] delta_log_w;                       // RW increments (symmetric)

  real<lower=0> omega;                                  // global clock-to-growth factor
}

transformed parameters {
  real<lower=0> delta_t = Tf_real - t0_real;
  array[Nevents] real<lower=0, upper=1> tevents;

  vector[Nnodes] log_w_node;
  vector<lower=0>[Nnodes] w_node;
  vector<lower=0>[Nnodes] nu_node;

  vector<lower=0>[Nnodes] terminal_clone_mass;
  simplex[Nnodes] clone_prob_cells;

  // --- growth: random walk on log w along the tree ---
  log_w_node[1] = log(w_bar);
  for (node in 2:Nnodes) {
    log_w_node[node] = log_w_node[parents[node]] + delta_log_w[node - 1];
  }
  w_node = exp(log_w_node);

  // --- clock: rigid coupling nu = omega * w (no residual) ---
  // computed in log space + clamped to avoid Inf*0 = NaN during warmup
  {
    real log_omega = log(omega);
    for (node in 1:Nnodes) {
      nu_node[node] = exp(fmin(log_omega + log_w_node[node], 50));
    }
  }

  // --- event times: nested-uniform stick-breaking ---
  for (idx in 1:Nevents) tevents[idx] = 0;
  for (node in 1:Nnodes) {
    int idx0 = idx_first_event[node];
    int p = parents[node];
    for (event in 1:number_of_events[node]) {
      int idx = idx0 + event - 1;
      if (p == 0) {
        tevents[idx] = (event == 1) ? u[idx] : u[idx] * tevents[idx0];
      } else {
        real t_parent = tevents[idx_first_event[p]];
```

```
        tevents[idx] = (event == 1)
          ? t_parent + u[idx] * (1.0 - t_parent)
          : t_parent + u[idx] * (tevents[idx0] - t_parent);
      }
    }
  }

  // --- abundance: exponential mass minus the children's contribution ---
  for (node in 1:Nnodes) {
    real t_node = tevents[idx_first_event[node]];
    terminal_clone_mass[node] = exp(fmin(w_node[node] * delta_t * (1.0 - t_node), 200));
  }
  for (child in 2:Nnodes) {
    int p = parents[child];
    real t_child = tevents[idx_first_event[child]];
    terminal_clone_mass[p] -= exp(fmin(w_node[p] * delta_t * (1.0 - t_child), 200));
  }
  for (node in 1:Nnodes) {
    if (terminal_clone_mass[node] < 1e-12) terminal_clone_mass[node] = 1e-12;
  }
  clone_prob_cells = terminal_clone_mass / sum(terminal_clone_mass);
}

model {
  // priors
  w_bar ~ lognormal(log(w_bar_prior_mean), w_bar_prior_sd);
  sigma_log_w ~ exponential(sigma_log_w_rate);
  delta_log_w ~ normal(0, sigma_log_w);
  omega ~ lognormal(log(omega_prior_mean), omega_prior_sd);
  // u ~ Uniform(0,1) implicit

  // mutation likelihood (Poisson), rate proportional to nu = omega * w
  for (node in 1:Nnodes) {
    int idx0 = idx_first_event[node];
    int p = parents[node];
    real t_p = (p == 0) ? 0.0 : tevents[idx_first_event[p]];

    for (event in 1:number_of_events[node]) {
      int idx = idx0 + event - 1;
      if (event == 1) {                                // MRCA -> diploid clonal m
        int idx_m = idx_first_mutation[idx0];
        real rate_m = nu_node[node] * mu * delta_t * lengths[idx0]
                      * coeff[idx_m] * (tevents[idx0] - t_p);
```

```
        target += poisson_lpmf(mutations[idx_m] | positive_rate(rate_m));
      } else {                                          // CNA -> alpha (pre) + beta (post)
        int idx_a = idx_first_mutation[idx];
        int idx_b = idx_first_mutation[idx] + 1;
        real t_cna = tevents[idx];
        real rate_a = nu_node[node] * mu * delta_t * lengths[idx]
                      * coeff[idx_a] * (t_cna - t_p);
        real rate_b = nu_node[node] * mu * delta_t * lengths[idx]
                      * coeff[idx_b] * (tevents[idx0] - t_cna);
        target += poisson_lpmf(mutations[idx_a] | positive_rate(rate_a));
        target += poisson_lpmf(mutations[idx_b] | positive_rate(rate_b));
      }
    }
  }

  // abundance likelihood (Negative Binomial)
  for (node in 1:Nnodes) {
    target += neg_binomial_2_lpmf(
      Ncells_per_clone[node] | positive_rate(Ncells * clone_prob_cells[node]),
      positive_rate(phi)
    );
  }
}

generated quantities {
  real w = w_bar;
  real omega_out = omega;
  vector[Nnodes] w_node_out = w_node;
  vector[Nnodes] nu_node_out = nu_node;
  vector[Nnodes] clone_prob_cells_out = clone_prob_cells;

  // per-node RW increment (drop-in with the simple model's output interface)
  vector[Nnodes] delta_log_w_out;
  delta_log_w_out[1] = 0;
  for (node in 2:Nnodes) {
    delta_log_w_out[node] = log_w_node[node] - log_w_node[parents[node]];
  }

  // posterior predictive replicated mutation counts (PPC on mutation numbers)
  array[Nequations] int mutations_rep;
  for (node in 1:Nnodes) {
    int idx0 = idx_first_event[node];
    int p = parents[node];
```

```
    real t_p = (p == 0) ? 0.0 : tevents[idx_first_event[p]];
    for (event in 1:number_of_events[node]) {
      int idx = idx0 + event - 1;
      if (event == 1) {
        int idx_m = idx_first_mutation[idx0];
        mutations_rep[idx_m] = poisson_rng(positive_rate(
          nu_node[node] * mu * delta_t * lengths[idx0] * coeff[idx_m] * (tevents[idx0] - t_p)));
      } else {
        int idx_a = idx_first_mutation[idx];
        int idx_b = idx_first_mutation[idx] + 1;
        real t_cna = tevents[idx];
        mutations_rep[idx_a] = poisson_rng(positive_rate(
          nu_node[node] * mu * delta_t * lengths[idx] * coeff[idx_a] * (t_cna - t_p)));
        mutations_rep[idx_b] = poisson_rng(positive_rate(
          nu_node[node] * mu * delta_t * lengths[idx] * coeff[idx_b] * (tevents[idx0] - t_cna)));
      }
    }
  }
}
```

# B. Appendix - Mathematical derivations

## B.1 System of Equations for Known Growth Rates

This appendix derives the system of equations that relates observed mutation counts to unknown event times under the assumption that the growth rate $w$ is known and constant across all clones. We present the derivation for a general branch and then illustrate the full system for a concrete example with four clones in a linear topology, where two copy-number alteration events occur on the branches of clones 2 and 3 respectively.

### General Derivation

For a branch without copy-number alterations, the expected number of mutations accumulated along the branch leading to clone $i$ is

$$\mathbb{E}[m_i] = \mu \cdot w \cdot l_{\text{tot},i} \cdot (t_i - t_{\text{pa}(i)}). \tag{B.1}$$

Treating the observed mutation count $m_i$ as equal to its expectation yields the deterministic equation

$$m_i = \mu \cdot w \cdot l_{\text{tot},i} \cdot (t_i - t_{\text{pa}(i)}), \tag{B.2}$$

which can be solved for $t_i$ given $t_{\text{pa}(i)}$:

$$t_i = \frac{m_i}{\mu \cdot w \cdot l_{\text{tot},i}} + t_{\text{pa}(i)}. \tag{B.3}$$

For a branch carrying a copy-number alteration event at time $t_{CN,ij}$, the mutations in the affected region are split into pre-alteration mutations $\alpha_{ij}$ and post-alteration mutations $\beta_{ij}$, yielding two equations:

$$\alpha_{ij} = \mu \cdot w \cdot l_{CN\text{tot},ij} \cdot c_{ij}^{\alpha} \cdot (t_{CN,ij} - t_{\text{pa}(i)}), \tag{B.4}$$

$$\beta_{ij} = \mu \cdot w \cdot l_{CN\text{tot},ij} \cdot c_{ij}^{\beta} \cdot (t_i - t_{CN,ij}). \tag{B.5}$$

Solving the first equation for $t_{CN,ij}$ gives

$$t_{CN,ij} = \frac{\alpha_{ij}}{\mu \cdot w \cdot l_{CN\text{tot},ij} \cdot c_{ij}^{\alpha}} + t_{\text{pa}(i)}, \tag{B.6}$$

and substituting into the second equation gives

$$t_i = \frac{\beta_{ij}}{\mu \cdot w \cdot l_{CN\text{tot},ij} \cdot c_{ij}^{\beta}} + t_{CN,ij}. \tag{B.7}$$

Together, these equations form a linear system that can be solved sequentially along the tree, starting from the root and proceeding toward the leaves.

## Example: Four Clones with Two Copy-Number Alterations

Consider a tumour with four clones arranged in a linear topology,

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4, \tag{B.8}$$

with $t_0 = 0$ and $T_f = 1$. Clone 2 carries a copy-number alteration event with timing $t_{CN,2}$, and clone 3 carries a copy-number alteration event with timing $t_{CN,3}$. The branches of clones 1 and 4 have no copy-number alterations. The full system of equations is as follows.

For the branch of clone 1, which has no copy-number alteration:

$$m_1 = \mu \cdot w \cdot l_1 \cdot t_1 \quad \longrightarrow \quad t_1 = \frac{m_1}{\mu \cdot w \cdot l_1}. \tag{B.9}$$

For the branch of clone 2, which carries a copy-number alteration at $t_{CN,2}$:

$$\alpha_2 = \mu \cdot w \cdot l_{CN,2} \cdot c_2^{\alpha} \cdot (t_{CN,2} - t_1) \quad \longrightarrow \quad t_{CN,2} = \frac{\alpha_2}{\mu \cdot w \cdot l_{CN,2} \cdot c_2^{\alpha}} + t_1, \tag{B.10}$$

$$\beta_2 = \mu \cdot w \cdot l_{CN,2} \cdot c_2^{\beta} \cdot (t_2 - t_{CN,2}) \quad \longrightarrow \quad t_2 = \frac{\beta_2}{\mu \cdot w \cdot l_{CN,2} \cdot c_2^{\beta}} + t_{CN,2}. \tag{B.11}$$

For the branch of clone 3, which carries a copy-number alteration at $t_{CN,3}$:

$$\alpha_3 = \mu \cdot w \cdot l_{CN,3} \cdot c_3^{\alpha} \cdot (t_{CN,3} - t_2) \quad \longrightarrow \quad t_{CN,3} = \frac{\alpha_3}{\mu \cdot w \cdot l_{CN,3} \cdot c_3^{\alpha}} + t_2, \tag{B.12}$$

$$\beta_3 = \mu \cdot w \cdot l_{CN,3} \cdot c_3^{\beta} \cdot (t_3 - t_{CN,3}) \quad \longrightarrow \quad t_3 = \frac{\beta_3}{\mu \cdot w \cdot l_{CN,3} \cdot c_3^{\beta}} + t_{CN,3}. \tag{B.13}$$

For the branch of clone 4, which has no copy-number alteration:

$$m_4 = \mu \cdot w \cdot l_4 \cdot (t_4 - t_3) \quad \longrightarrow \quad t_4 = \frac{m_4}{\mu \cdot w \cdot l_4} + t_3. \tag{B.14}$$

The system can be solved sequentially: given the observed mutation counts $m_1$, $\alpha_2$, $\beta_2$, $\alpha_3$, $\beta_3$, $m_4$, the known mutation rate $\mu$, growth rate $w$, and genomic lengths $l$, all event times $t_1, t_{CN,2}, t_2, t_{CN,3}, t_3, t_4$ can be recovered exactly. The Poisson likelihoods in the main model represent the probabilistic counterpart of this deterministic system, where mutation counts are treated as noisy observations of their expected values.

## B.2 System of Equations for Unknown Growth Rates

When the growth rates are unknown, the deterministic system described in Appendix B.1 is no longer solvable in closed form, since each equation now contains two unknowns: the event time and the clone-specific growth rate $w_i$. Specifically, for a branch without copy-number alterations,

$$m_i = \mu \cdot w_i \cdot l_{\text{tot},i} \cdot (t_i - t_{\text{pa}(i)}), \tag{B.15}$$

and only the product $w_i \cdot (t_i - t_{\text{pa}(i)})$ is identified by the data. Similarly, for branches with copy-number alterations,

$$\alpha_{ij} = \mu \cdot w_i \cdot l_{CN\text{tot},ij} \cdot c_{ij}^{\alpha} \cdot (t_{CN,ij} - t_{\text{pa}(i)}), \tag{B.16}$$

$$\beta_{ij} = \mu \cdot w_i \cdot l_{CN\text{tot},ij} \cdot c_{ij}^{\beta} \cdot (t_i - t_{CN,ij}), \tag{B.17}$$

where the clone growth rate $w_i$ is now itself unknown, so that each equation mixes an event time with a growth rate. The system is therefore underdetermined, and additional information is needed to separate timings from growth rates.

### Example: Four Clones with Two Copy-Number Alterations and Unknown Growth Rates

Consider the same four-clone linear topology as in Appendix B.1,

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4, \tag{B.18}$$

but now with clone-specific growth rates $w_1, w_2, w_3, w_4$ treated as unknown. Both the pre- and post-alteration mutations on the branch of clone $i$ lie on the stem lineage of that clone and therefore accumulate at its own rate $w_i$; the split at $t_{CN,ij}$ changes only the number of allelic copies available for new mutations, through $c_{ij}^{\alpha}$ and $c_{ij}^{\beta}$, not the rate. The full underdetermined system is:

$$m_1 = \mu \cdot w_1 \cdot l_1 \cdot t_1, \tag{B.19}$$

$$\alpha_2 = \mu \cdot w_2 \cdot l_{CN,2} \cdot c_2^{\alpha} \cdot (t_{CN,2} - t_1), \tag{B.20}$$

$$\beta_2 = \mu \cdot w_2 \cdot l_{CN,2} \cdot c_2^{\beta} \cdot (t_2 - t_{CN,2}), \tag{B.21}$$

$$\alpha_3 = \mu \cdot w_3 \cdot l_{CN,3} \cdot c_3^{\alpha} \cdot (t_{CN,3} - t_2), \tag{B.22}$$

$$\beta_3 = \mu \cdot w_3 \cdot l_{CN,3} \cdot c_3^{\beta} \cdot (t_3 - t_{CN,3}), \tag{B.23}$$

$$m_4 = \mu \cdot w_4 \cdot l_4 \cdot (t_4 - t_3). \tag{B.24}$$

This system has 6 equations in 10 unknowns: the six event times $t_1, t_{CN,2}, t_2, t_{CN,3}, t_3, t_4$ and the four growth rates $w_1, w_2, w_3, w_4$. It is therefore underdetermined by 4 degrees of freedom, and no unique solution exists from the mutation data alone.

## The Abundance Constraint and the Residual Scale

The count above uses only the mutation data. The model, however, also observes the clone abundances, and these add further constraints. Suppose first that the *absolute* size of each clade were observed. The terminal mass of clone $i$ would then satisfy

$$N_i = e^{\,w_i\,(T_f - t_i)} \quad \Longrightarrow \quad \ln N_i = w_i\,(T_f - t_i), \tag{B.25}$$

one equation per clone. Combined with the mutation equation $m_i = \mu \cdot w_i \cdot l_i \cdot (t_i - t_{\mathrm{pa}(i)})$, the two time intervals telescope,

$$\frac{m_i}{\mu\, l_i} + \ln N_i = w_i\,(t_i - t_{\mathrm{pa}(i)}) + w_i\,(T_f - t_i) = w_i\,(T_f - t_{\mathrm{pa}(i)}), \tag{B.26}$$

which solves for the growth rate and then for the event time,

$$w_i = \frac{m_i/(\mu\, l_i) + \ln N_i}{T_f - t_{\mathrm{pa}(i)}}, \qquad t_i = t_{\mathrm{pa}(i)} + \frac{m_i}{\mu\, l_i\, w_i}. \tag{B.27}$$

With absolute clade sizes the system would therefore be *exactly identified*: each clone contributes one mutation equation and one abundance equation, matching its two unknowns $w_i$ and $t_i$ (on a copy-number branch the extra equation $\beta_{ij}$ matches the extra unknown $t_{CN,ij}$).

The single-cell data, however, do not provide absolute sizes. Sequencing reads a number of cells $N_{\text{tot}}$ fixed by the experiment, not the true tumour size, so what is observed is

$$N_i = c \cdot e^{\,w_i\,(T_f - t_i)}, \qquad c = \frac{N_{\text{tot}}}{\sum_k M_k}, \tag{B.28}$$

with an unknown common sampling factor $c$; equivalently, only the proportions $p_i = M_i / \sum_k M_k$ are measured. Taking logarithms,

$$\ln N_i = w_i\,(T_f - t_i) + \ln c, \tag{B.29}$$

so the abundances determine $w_i\,(T_f - t_i)$ only up to the common constant $\ln c$: they fix the $K-1$ *ratios* of clade sizes but not their overall scale. In the four-clone example these three independent ratios cut the under-determination from four down to one; in general the $K$ mutation equations and the $K-1$ abundance ratios pin the $2K$ growth-and-time unknowns up to a single residual degree of freedom (each copy-number branch adds one timing unknown and one equation, leaving the balance unchanged). That one remaining direction is precisely the global growth scale $c$, which a single snapshot cannot resolve because the composition of one sample is invariant to it. It is this residual scale, rather than four free directions, that the hierarchical prior must fix, as discussed next; recovering it from the data instead would require an absolute measure of tumour burden, ideally collected at more than one time.

## Hierarchical Prior as Regularisation

The additional information needed to regularise the system is provided by the hierarchical prior on the growth rates described in Section 2.2. The random walk on the log scale assigns

$$\log w_1 = \log \bar{w}, \tag{B.30}$$

$$\log w_2 = \log w_1 + \sigma_{\log w} \cdot z_2, \tag{B.31}$$

$$\log w_3 = \log w_2 + \sigma_{\log w} \cdot z_3, \tag{B.32}$$

$$\log w_4 = \log w_3 + \sigma_{\log w} \cdot z_4, \tag{B.33}$$

with $z_2, z_3, z_4 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ and $\log \bar{w} \sim \mathcal{N}(\log \bar{w}_{\text{prior}}, \sigma_{\bar{w}})$. This introduces 4 additional parameters $(\bar{w}, z_2, z_3, z_4)$ governed by prior distributions that constrain their values, effectively regularising the posterior. The prior encodes the biological assumption that related clones tend to have similar growth rates: the further two clones are apart in the tree, the more their rates are allowed to differ, since deviations accumulate along the branches.

In the limit $\sigma_{\log w} \to 0$, all growth rates collapse to $\bar{w}$ and the system reduces to the identifiable case of Appendix B.1, where $w = \bar{w}$ is known. For finite $\sigma_{\log w}$, the prior allows the clone rates to differ while penalising large departures.

It is worth being precise about what this regularisation does and does not achieve. It makes the posterior *proper* and lets the sampler explore plausible combinations of timings and growth rates, but it does not make the growth rates *identified by the data*. Along the single residual direction identified in the previous subsection, the global growth scale $c$, both the mutation likelihood and the abundance proportions are unchanged, so there the posterior simply follows the prior. The residual uncertainty this leaves on the per-clone rates is exactly the single-snapshot floor observed in the simulation study, and it cannot be removed by sharpening the prior: a tighter prior would narrow the posterior only by asserting information the data do not provide, at the cost of calibration. What *is* learned from the data is the degree of pooling toward a common rate, through the posterior of $\sigma_{\log w}$; the overall growth scale, by contrast, is set by the prior rather than by the single snapshot.

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